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Advisor: Haochen Li
June 2026

Exploring pLM Ensembles

Analyzing how protein Language Models can be combined to improve downstream predictive performance

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Graduation thesis submitted in partial fulfillment of the requirements for the degree of de Ingenieurswetenschappen: Computerwetenschappen

EXPLORING PLM ENSEMBLES

Analyzing How Protein Language Models
Can Be Combined to Improve
Downstream Predictive Performance

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June 2026

Promotor: prof. dr. Wim Vranken

Advisor: Haochen Li
sciences and bioengineering sciences

Research Training

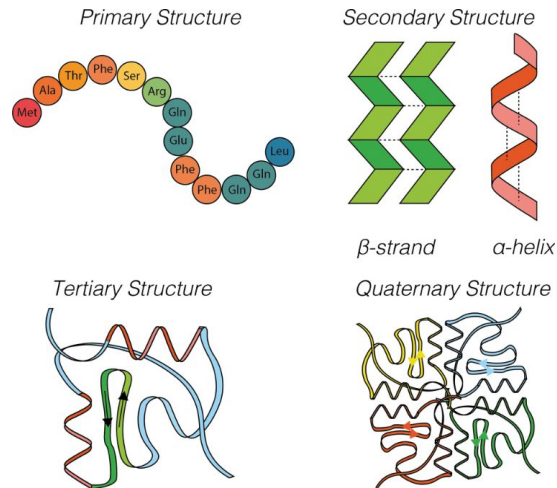
Background – Biology

Protein **function** defined by **conformation**

⇒ **Protein dynamics** = constant **microscopic motion**

Many types of mutations

⇒ **Mutations** may or may not **affect function**



(Morris R et al., 2022)

	No mutation	Silent mutation	Missense mutation	Nonsense mutation	Frameshift mutation
DNA	GAA TTC CGT	GAA TTT CGT	GAA TCC CGT	GAA ATC CGT	GAA AAT CCG
mRNA	CUU AAG GCA	CUU AAA GCA	CUU AGG GCA	CUU UAG GCA	CUU UUA GGC
protein	Glu Lys Gly	Glu Lys Gly	Glu Arg Gly	Glu STOP	Glu Asn Gly

(Morris R et al., 2022)

Research Training

Background – Machine Learning vs Deep Learning

Machine Learning

1. Data pre-processing
2. Feature extraction
3. Model training

- + Works with less data
- + More explainable
- Generally worse performance

Deep Learning

1. Data pre-processing
- ~~2. Feature extraction~~
3. Model training

- Needs more data to capture features
- Requires more sophisticated XAI techniques
- + Generally better performance

Research Training

Background – Protein Language Models (pLMs)

LMs pre-trained on protein data (e.g. sequences, evolutionary/functional/... data, ...)

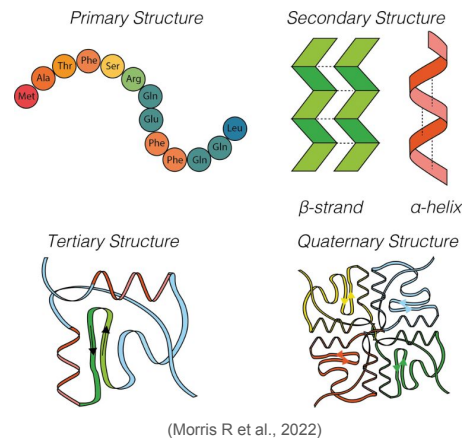
Captures **protein properties** given an **Amino Acid (AA) sequence**

Protein properties are defined by **conformation**

⇒ **conformation** is defined by **folds**

⇒ **folds** depend on the **AA sequence**

⇒ pLMs learn these implicitly during pre-training (using e.g. UniRef)



Research Training

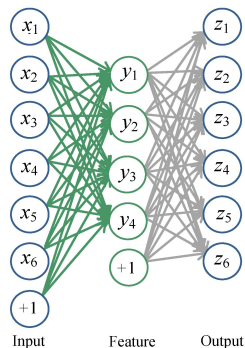
Background – Differences between pLMs

Many types of base architectures:

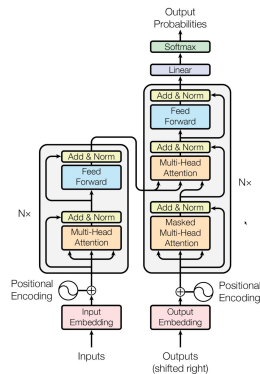
- **Transformers** $\Rightarrow$ parallel attention (BERT/T5/...)
- **RNNs** $\Rightarrow$ sequential attention (mLSTM)
- **CNNs** $\Rightarrow$ spatial attention (made for vision tasks)

Many types of learning objectives:

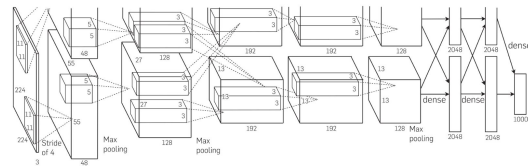
- Masked language modelling
- Autoregression
- Auto-encoding
- ...



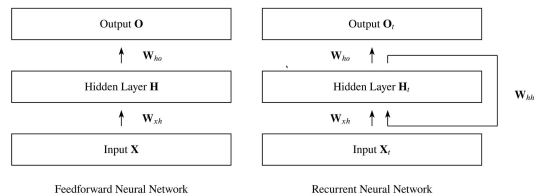
Auto-encoder (S Chen, W Guo, 2023)



Transformer (Vaswani A. et al., 2017)



CNN (Krizhevsky A. et al., 2012)



RNN (Robin M., 2019)

Research Training

Background – Ensemble Learning

Ensembles: aggregation of many model decisions/representations

⇒ “wisdom of crowds” principle

= Different biases of independent sources

Many taxonomies across the literature

Popular ensemble model: XGBoost

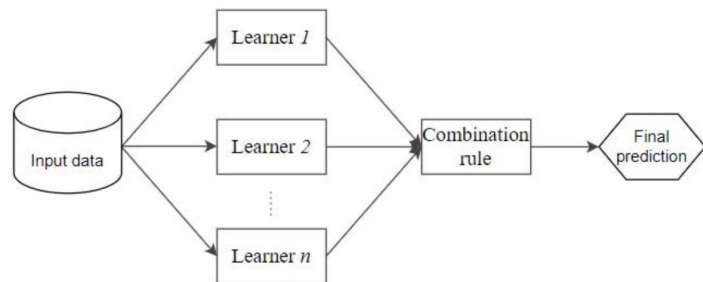


FIGURE 1. Block diagram of parallel ensemble learning.

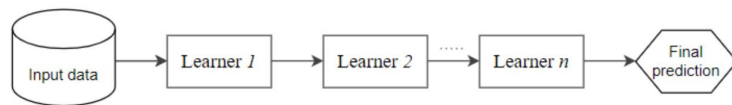


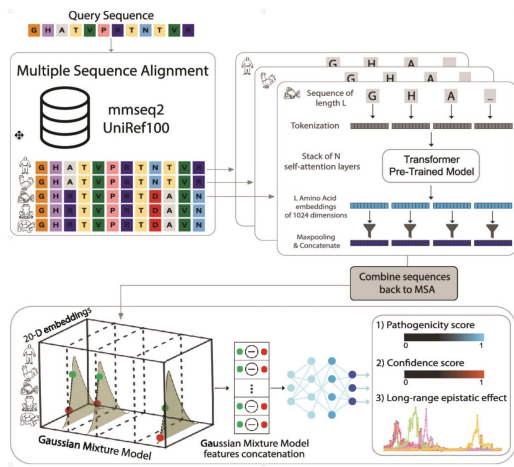
FIGURE 2. Block diagram of sequential ensemble learning.

Parallel vs Sequential Ensemble Learning (I. D. Mienye, Y. Sun, 2022)

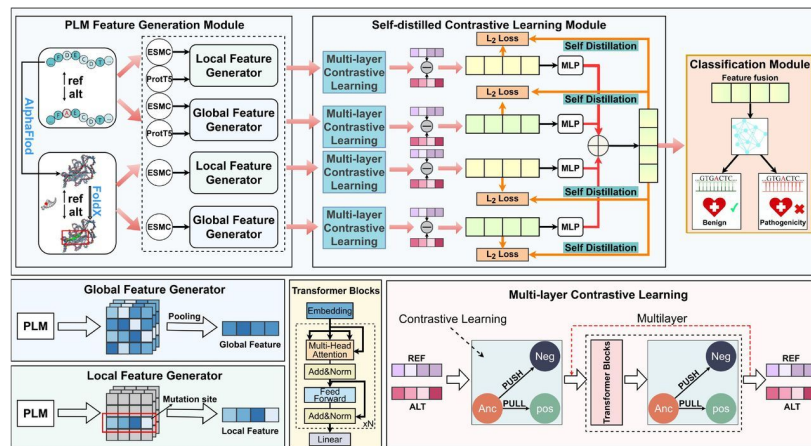
Research Training

Related Work

- pLM aggregate models: Venus-MAXWELL & D2Deep
- pLM ensembles: VarDCL



D2Deep (Tzavella K., 2025)



VarDCL (Zhang H., 2026)

Introduction

Problem Statement & Motivation

Interesting winning solution to Netflix Prize (Bellkor)

⇒ Uses an ensemble of the best performing models



(New York Times, 2009)

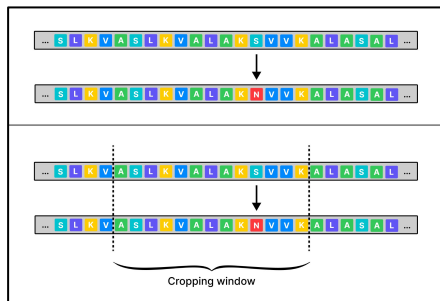
Goal: to better understand how different models behave when combined

⇒ Better understanding → better models/architectures

⇒ Advancements in e.g. drug development

Methods

Overview



Dataset & pre-processing

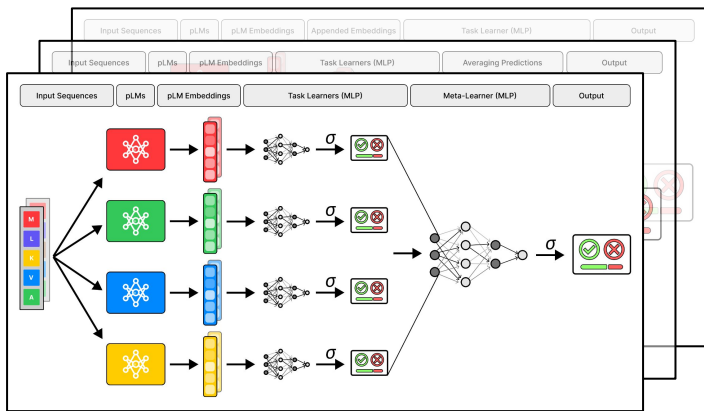
Algorithm 1: Embedding pre-computation pipeline pseudocode.

```

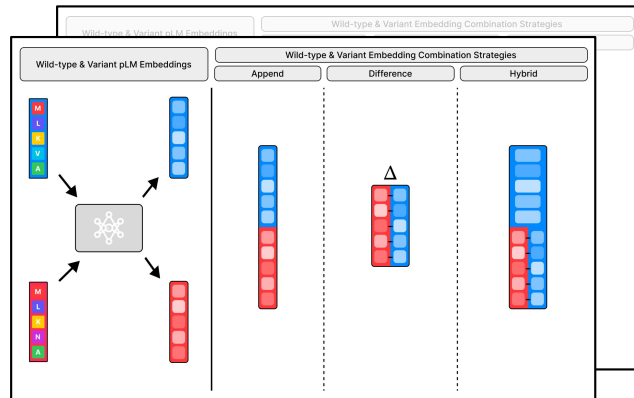
1 # Load the train and test datasets
2 dataset = load_dataset()
3
4 # Load the model into (video) memory
5 plm_model = load_plm(plm)
6
7 # Initialize an empty dictionary
8 embedding_dict = {}
9
10 # Extract and save embeddings for each sample
11 embeddings = plm_model(sample.sequence)
12 mut_embeddings = plm_model(sample.mut_sequence)
13
14 new_id = sample.id + "." + sample.variant
15 embedding_dict[new_id][0] = embeddings
16 embedding_dict[new_id][1] = mut_embeddings
17
18 # Persist the dictionary to disk
19 save_to_disk(embedding_dict)

```

Pipelines



Architectures



Experiments

Methods

Dataset – Cancer Driver Mutation Binary Classification

Samples: 3644/3644 samples (train/test)

⇒ (<protein_id>, <variant>, <target>)

⇒ fetching sequences from UniProt

Variant: <original_AA>, <subst_idx>, <new_AA>

⇒ e.g. S46N → S replaced by N at index 46

Balance: 45/55% (+/-)

⇒ Fairly balanced

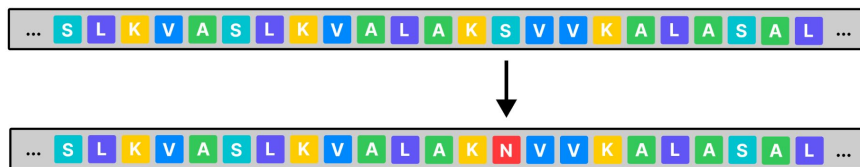
Methods

Protein Sequence Cropping

Some sequences are too long for most pLMs (token limits)

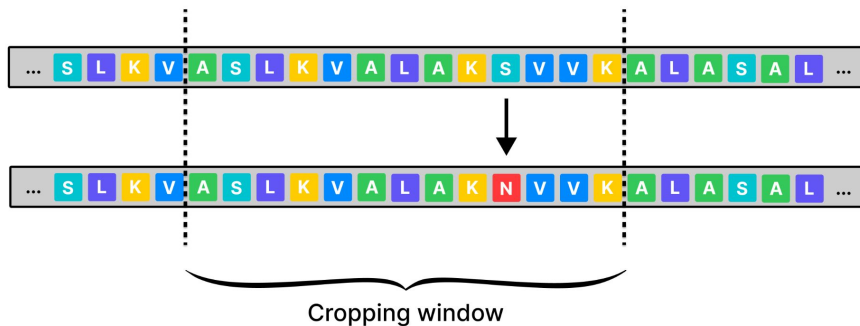
⇒ cropping sequences needed

⇒ around mutation site



Random relative mutation index

⇒ to avoid positional bias



Methods

Ensemble Architectures – Taxonomy

Feature-Level $\Leftrightarrow$ Decision-Level

- Feature-Level: Fusion **before** learners
⇒ early fusion
- Decision-Level: Fusion **after** learners
⇒ late fusion

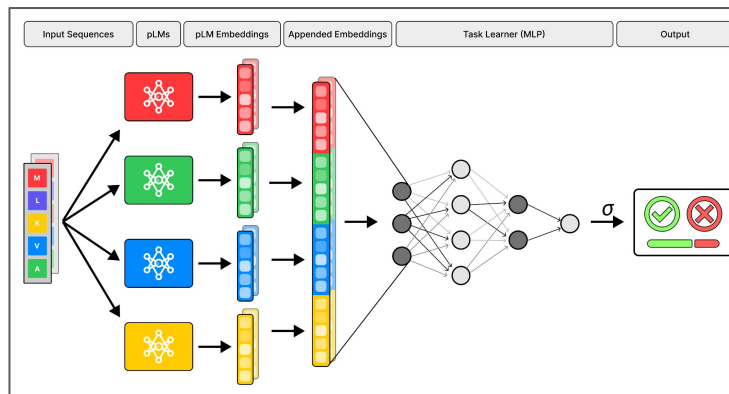
Parallel $\Leftrightarrow$ Layered

- Parallel: All learners **trained simultaneously**
- Layered: **Sequential training** of learners

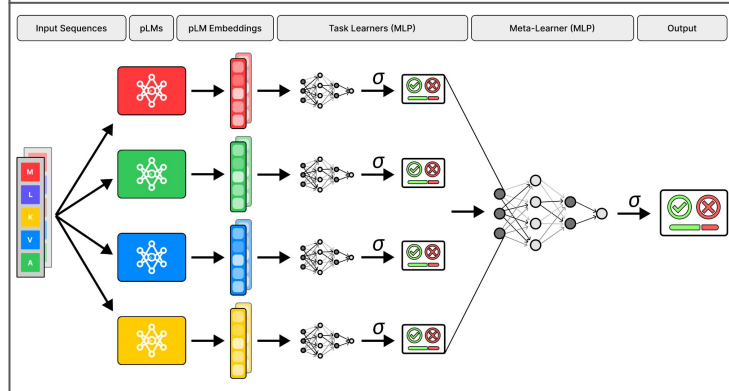
Methods

Ensemble Architectures

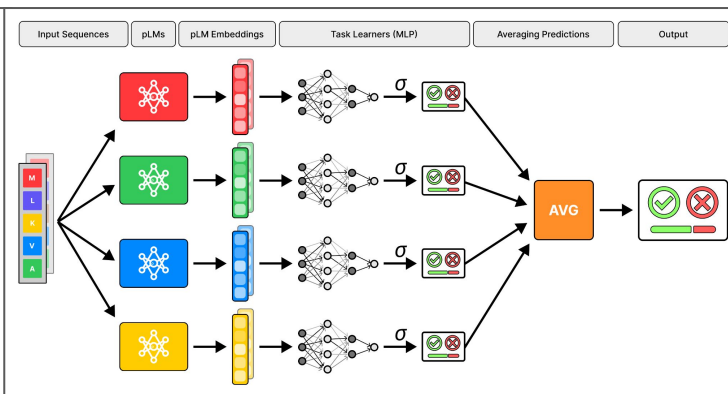
RF



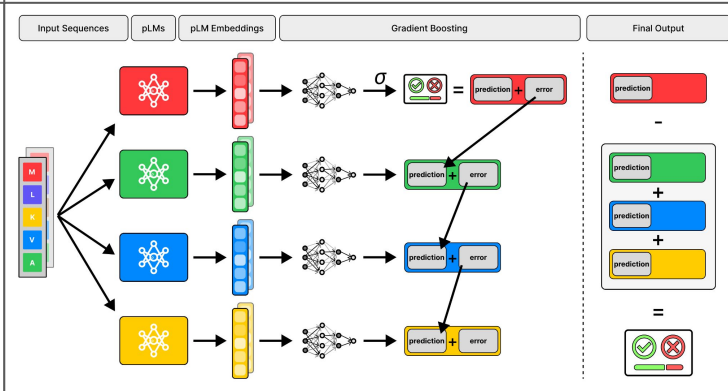
ML



TL

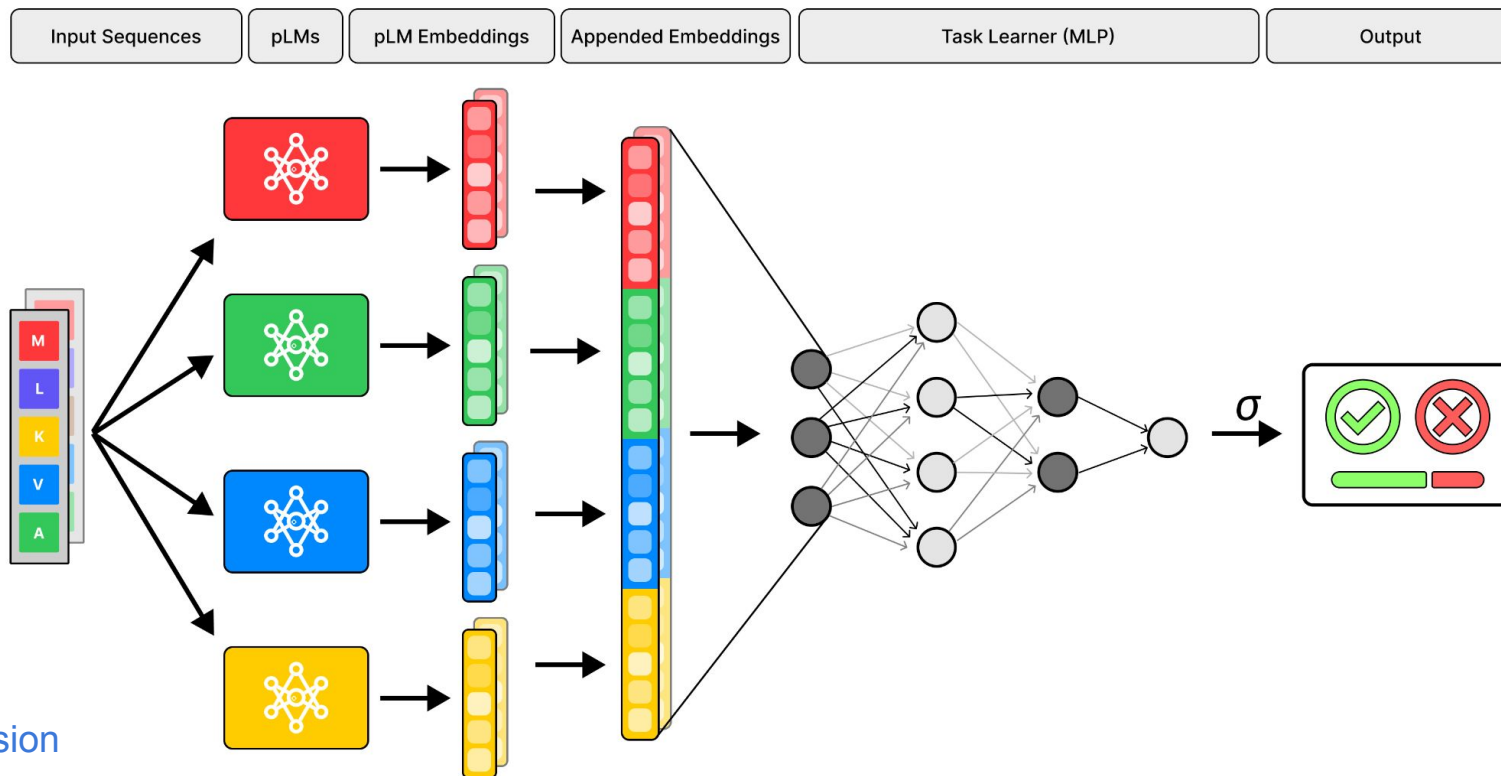
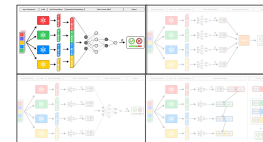


GB



Methods

Ensemble Architectures – Representation Fusion (RF)

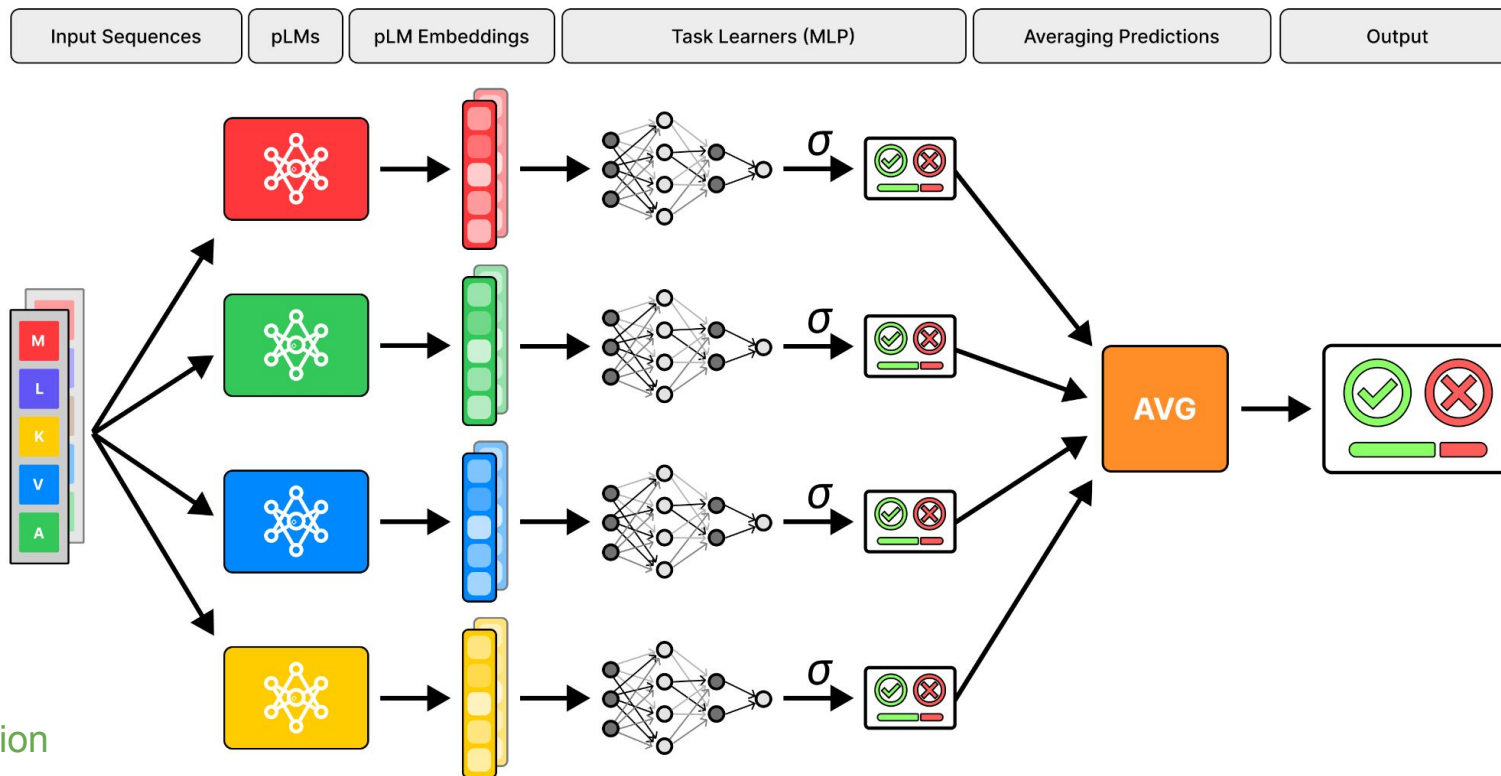
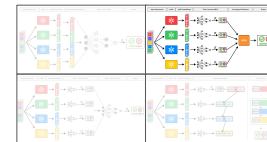


Early Fusion

Parallel

Methods

Ensemble Architectures – Task-Level (TL)

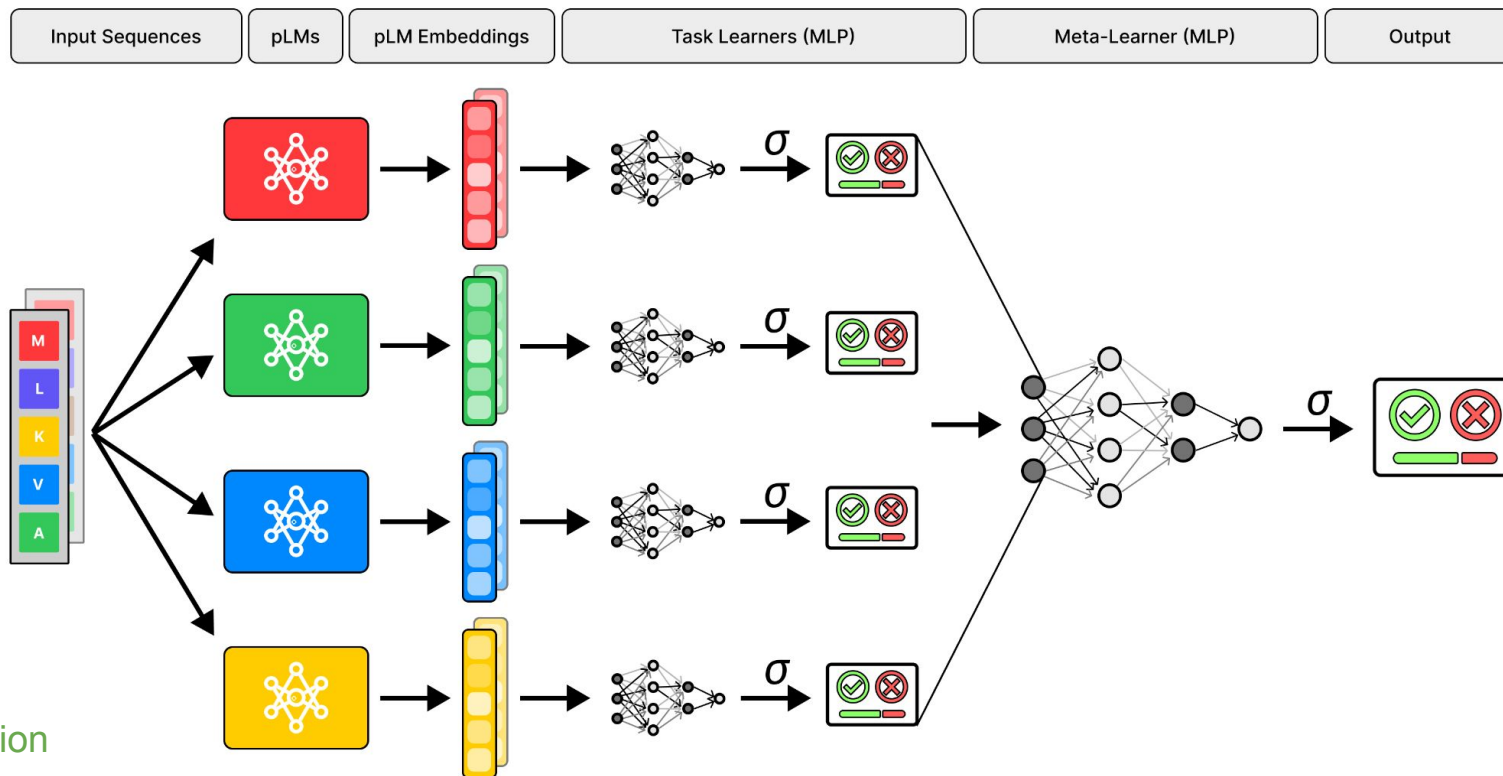
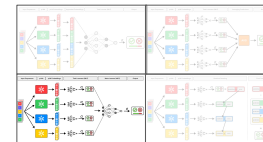


Late Fusion

Parallel

Methods

Ensemble Architectures – Task-Level with Meta-Learner (ML)

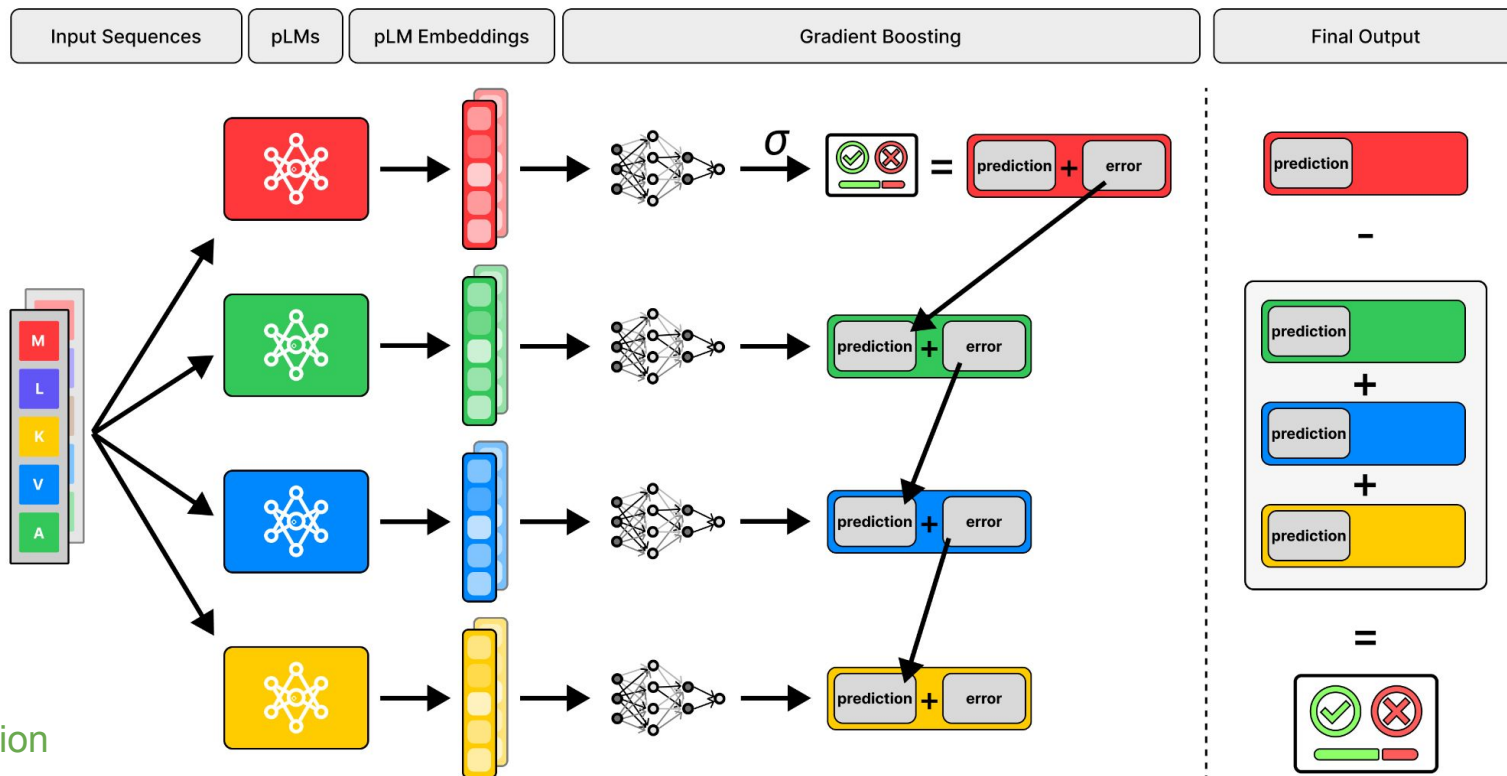
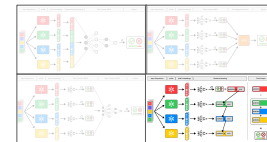


Late Fusion

Layered

Methods

Ensemble Architectures – Gradient Boosting (GB)



Late Fusion

Layered

Methods

Selected pLMs

- Largest pLM models publicly available
- 12 pLMs (different sizes, #embedding dimensions, base architecture, training objective, ...)

pLM	Version Used	Year	Parameters	Emb. Dim.	Architecture Type	Specialization
<i>General-Purpose Models</i>						
UniRep	UniRep-mLSTM	2019	18M	1900	RNN (mLSTM)	General-Purpose
ESM-1b	ESM-1b-650M	2021	650M	1280	Transformer (BERT)	General-Purpose
ProtT5	ProtT5-XL-U50	2021	3B	1024	Transformer (T5)	General-Purpose
ESM-2	ESM2-15B	2022	15.0B	5120	Transformer (BERT)	General-Purpose
ProGen2	ProGen2-XL	2022	6.4B	4096	Transformer (Autoregressive)	General-Purpose
CARP	CARP-640M	2023	640M	1280	CNN	General-Purpose
ESM-C	ESM-C-600M	2024	600M	1152	Transformer (BERT)	General-Purpose
Ankh3	Ankh3-large	2025	1.1B	1536	Transformer (T5)	General-Purpose
<i>Domain Specialized</i>						
AntiBERTy	AntiBERTy	2021	26M	512	Transformer (BERT)	Antibody Specialized
ESM-1v	ESM-1v-650M	2021	650M	1280	Transformer (BERT)	Variant Specialized
SeqDance	SeqDance	2025	35M	45	Transformer (BERT)	Protein Dynamics
ESMDance	ESMDance	2025	35M	45	Transformer (BERT)	Protein Dynamics

Methods

Pipelines

Embedding Pre-Computation

Algorithm 1: Embedding pre-computation pipeline pseudocode.

```
1 # Load the train and test datasets
2 dataset = load_dataset()

3 foreach pLM do
4     # Load the model into (video) memory
5     plm_model = load_plm(pLM)

6     # Initialize an empty dictionary
7     embedding_dict = {}

8     foreach sample in dataset do
9         # Extract and save embeddings for each sample
10        embeddings = plm_model(sample.sequence)
11        mut_embeddings = plm_model(sample.mut.sequence)

12        new_id = sample.id + "_" + sample.variant
13        embedding_dict[new_id][0] = embeddings
14        embedding_dict[new_id][1] = mut_embeddings

15    # Persist the dictionary to disk
16    save_to_disk(embedding_dict)
```

Ensemble Learning

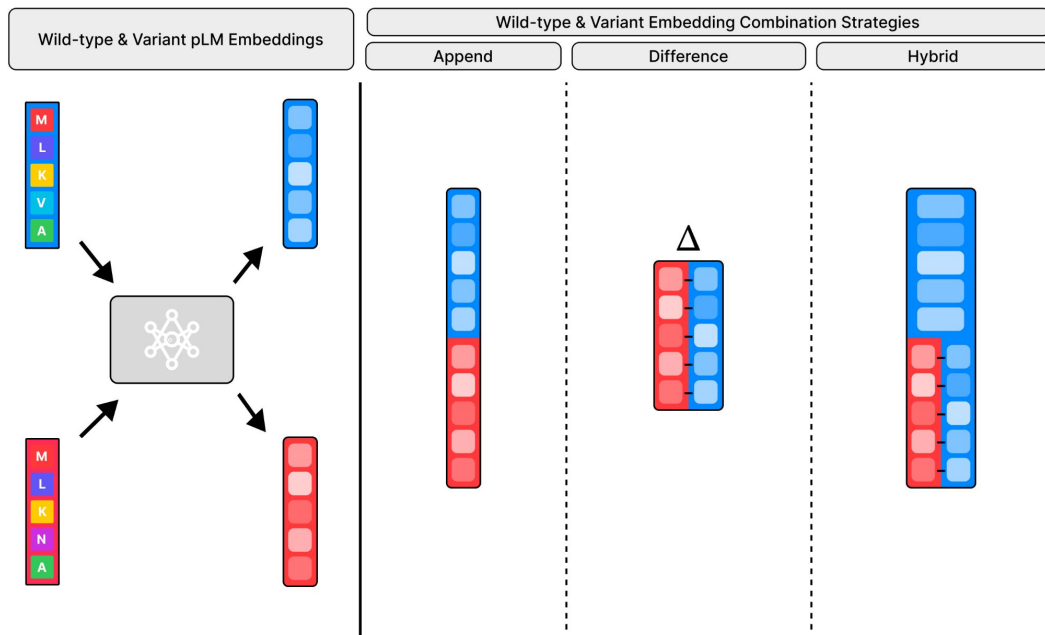
- Early-stopping with patience
- MLP backpropagation
- Stratified group K-fold cross-validation
 - Validate on unseen proteins
 - Keeping fold balance

Methods

Experiments

Experiment 1 – Wild-Type & Variant Combination Strategies

- Which **combination strategy** result in the best **predictive performance**?



Methods

Experiments

Experiment 1 – Wild-Type & Variant Combination Strategies

- Which **combination strategy** result in the best **predictive performance**?

Experiment 2 – Individual pLM Performance

- Which **pLMs** result in the best **predictive performance**?
- Which pLMs generates **meaningful latent features**?

Experiment 3 – Ensemble Architecture Performance

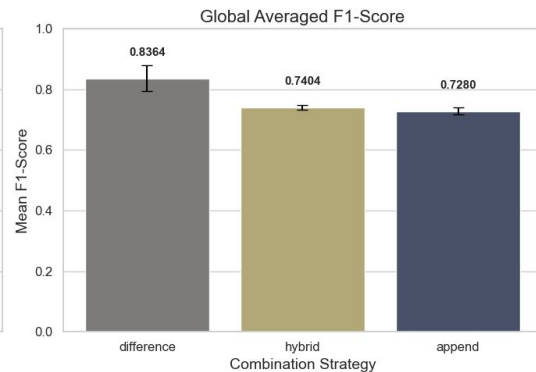
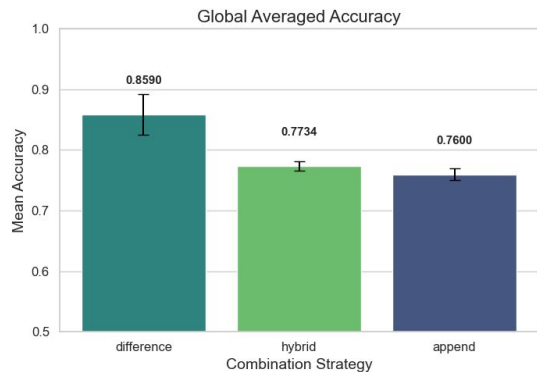
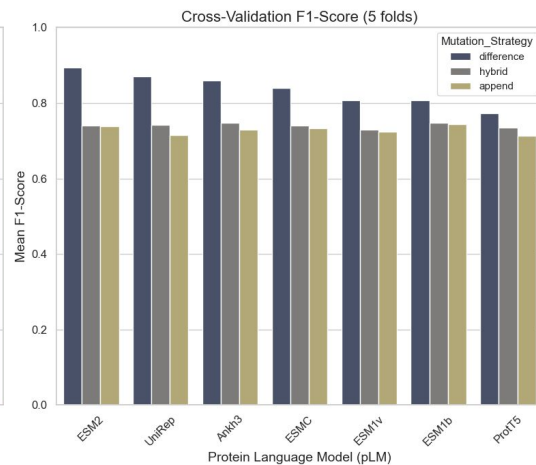
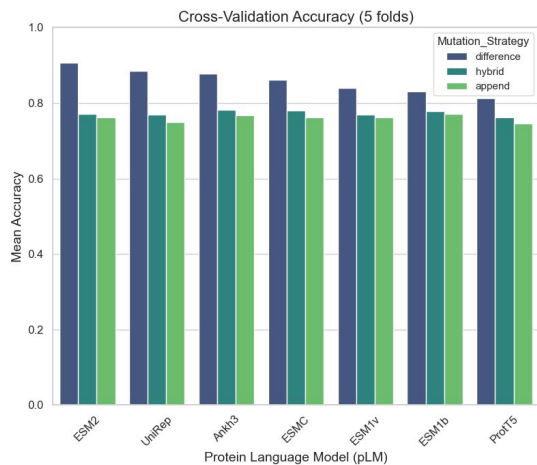
- Which **ensemble architecture** result in the best **predictive performance**?

Experiment 4 – Feature-Level pLM Combination Analysis (Feature Attribution)

- How do the pLMs' **latent spaces interact** with each other?

Results

Experiment 1 – Wild-Type & Variant Combination Strategies



Difference outperforms both

⇒ contains **explicit mutation information**

⇒ model **focuses on mutation**, not context

Append is the worst strategy

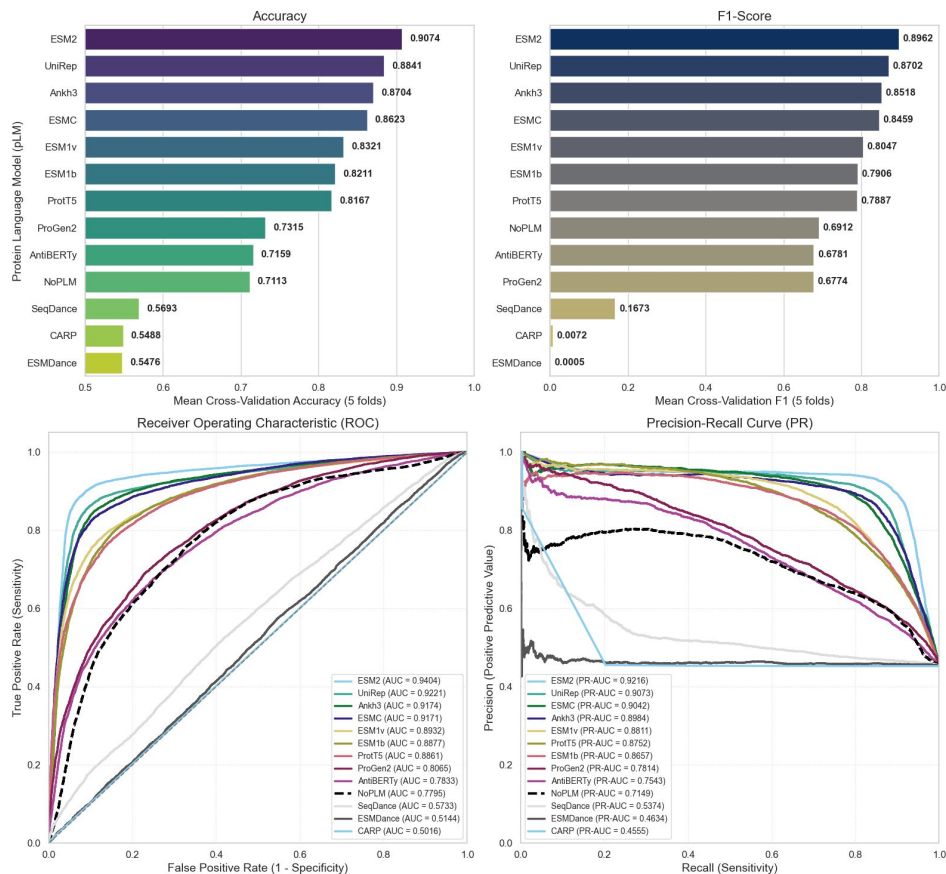
⇒ doubles input dimensionality

⇒ no explicit mutation information

Hybrid is slightly better than append

Results

Experiment 2 – Individual pLM Performance



Larger models > smaller models

⇒ Except:

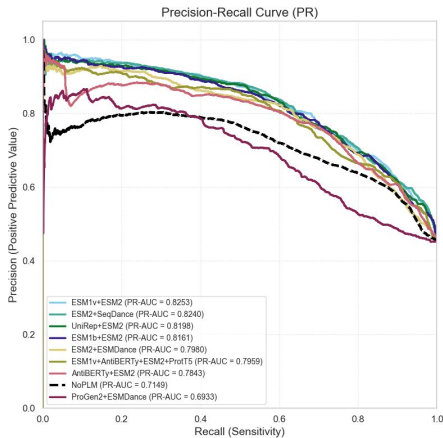
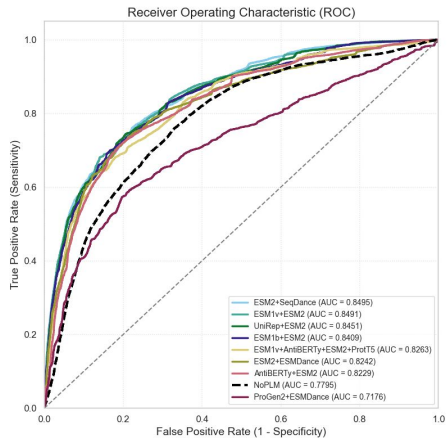
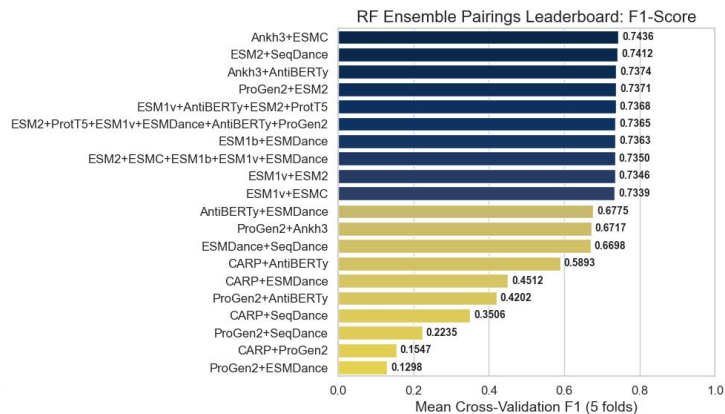
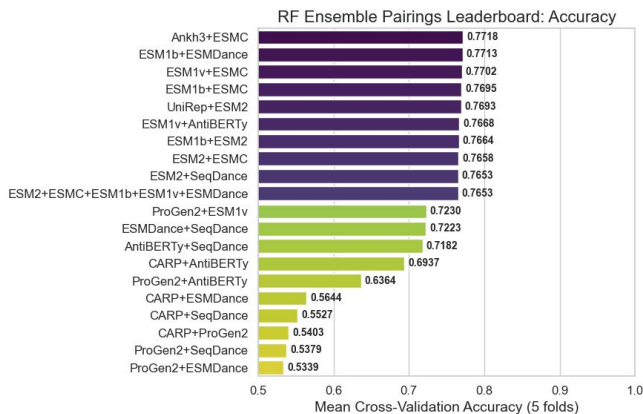
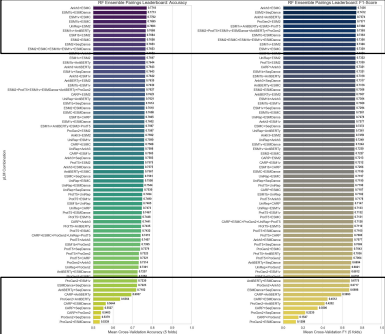
- ProGen2 (autoregressive)
- CARP (CNN)

SeqDance, CARP, ESMDance < NoPLM

⇒ ESM-2 outperforms all other pLMs

Results

Experiment 3 – Ensemble Architecture Performance – RF



Best Combination: Ankh3 + ESM-C

Worst combination: ProGen2 + ESMDance

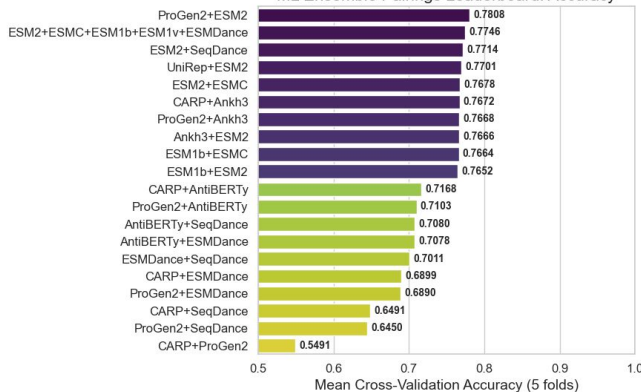
Interesting case: ESM-2 + SeqDance

⇒ Best AUC scores

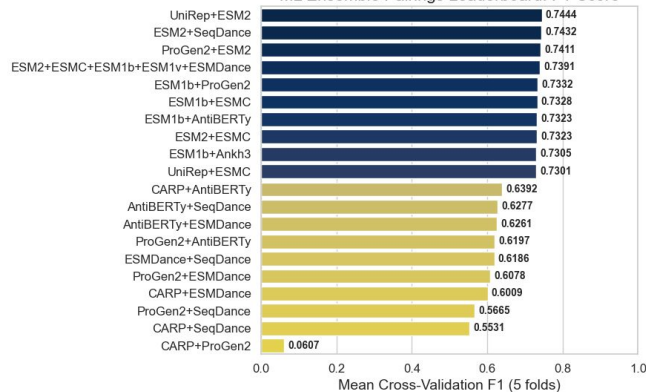
Results

Experiment 3 – Ensemble Architecture Performance – ML

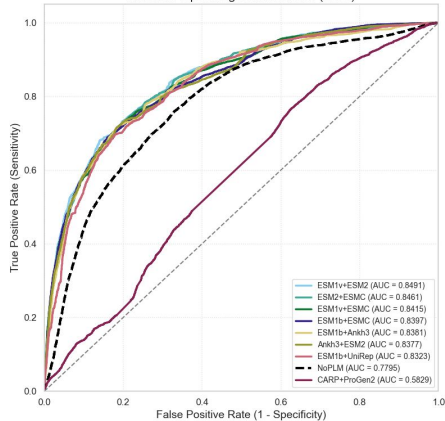
ML Ensemble Pairings Leaderboard: Accuracy



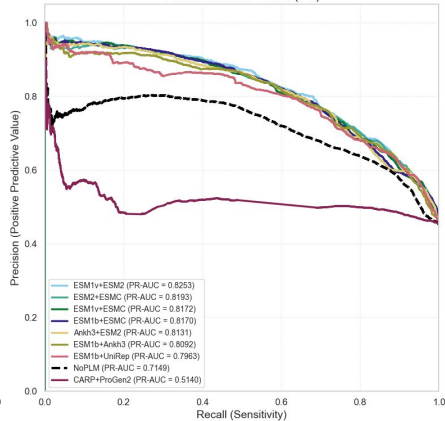
ML Ensemble Pairings Leaderboard: F1-Score



Receiver Operating Characteristic (ROC)



Precision-Recall Curve (PR)

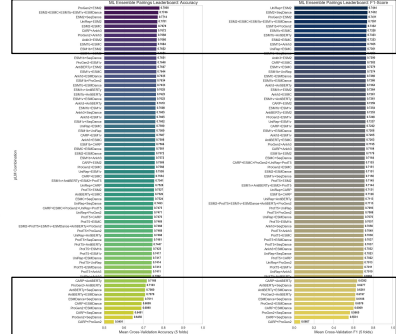


Best Combination: UniRep + ESM-2

Worst combination: CARP + ProGen2

⇒ Best accuracy/F1-score

(more expressive version of TL)



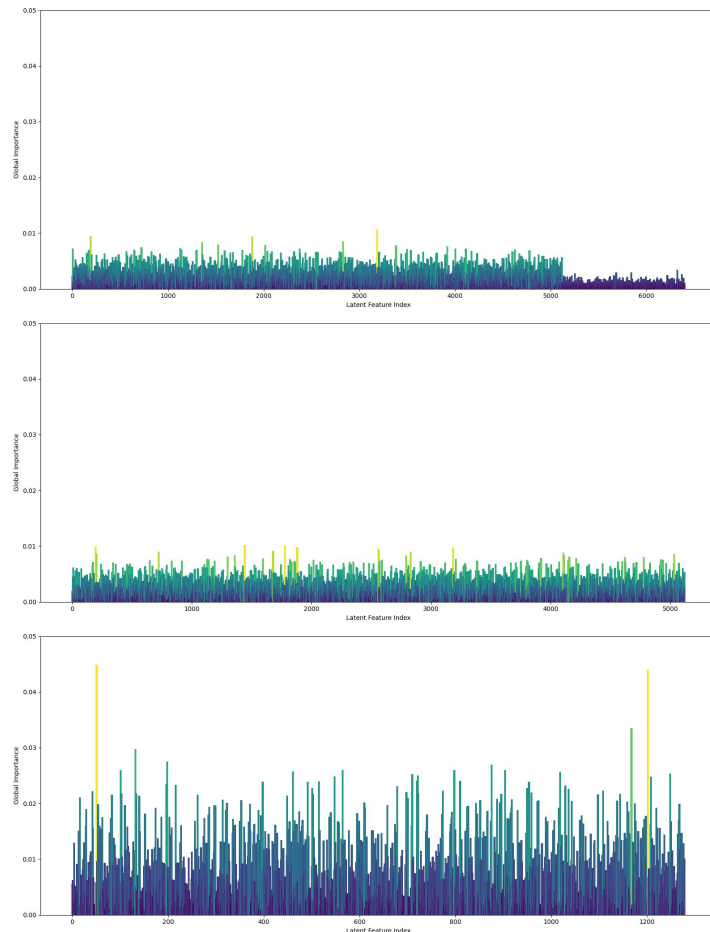
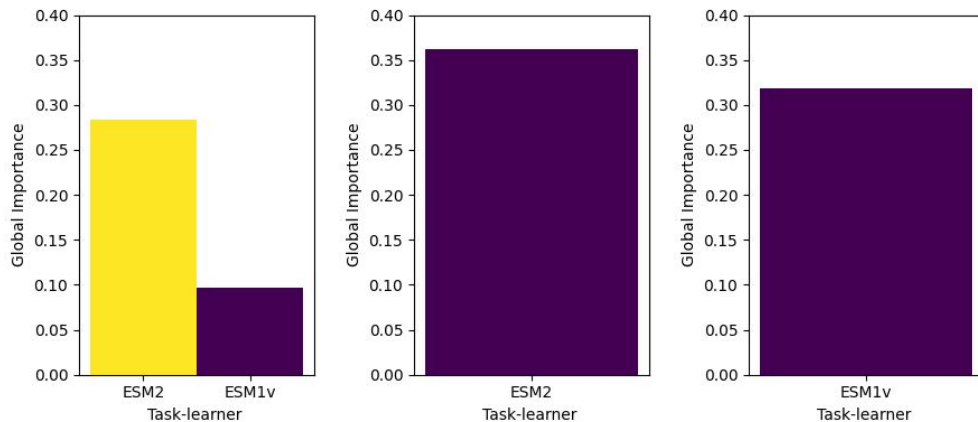
Results

Experiment 4 – Feature-Level pLM Combination Analysis

ESM-2 & ESM-1v

Change in score of important features

⇒ Overlapping latent spaces



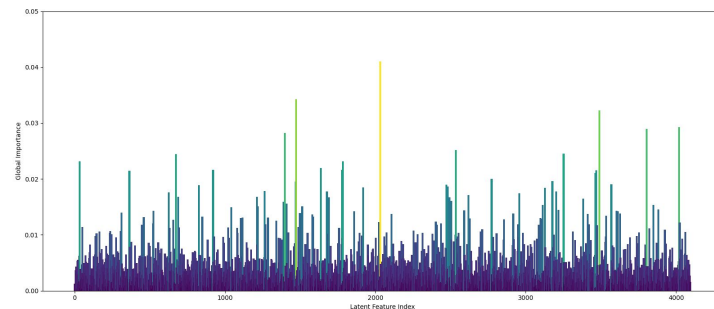
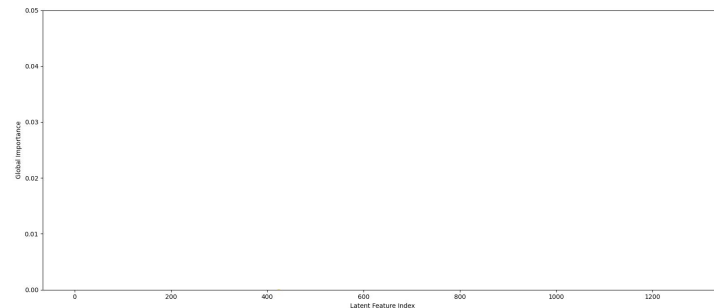
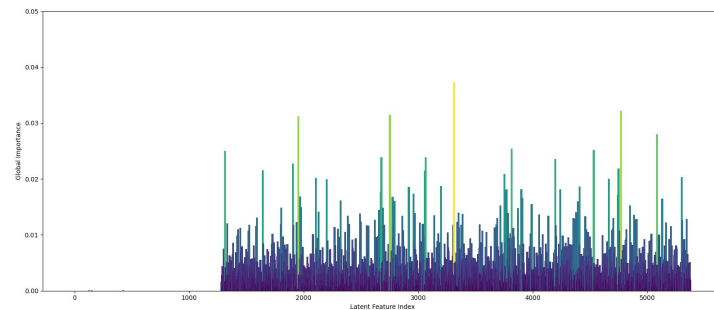
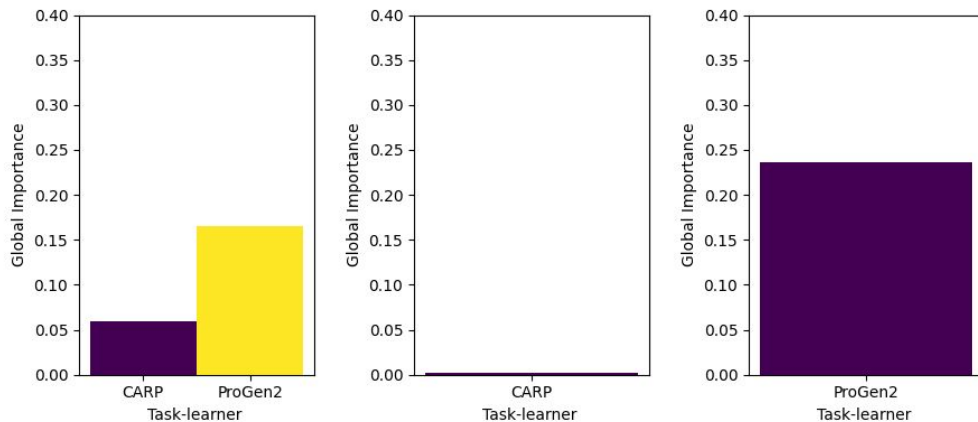
Results

Experiment 4 – Feature-Level pLM Combination Analysis

CARP & ProGen2

Exclusive reliance on ProGen2

⇒ Overshadowing latent spaces



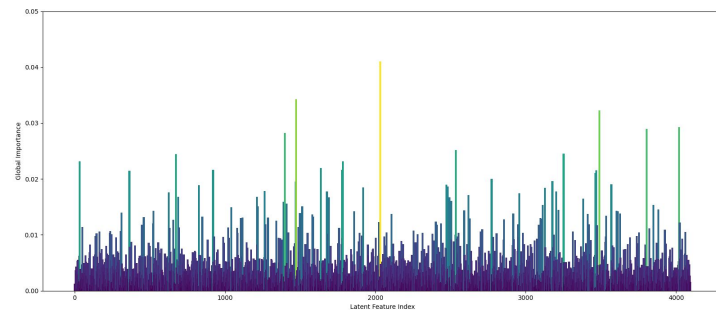
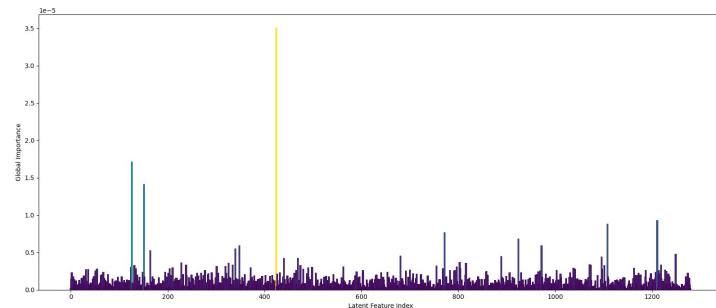
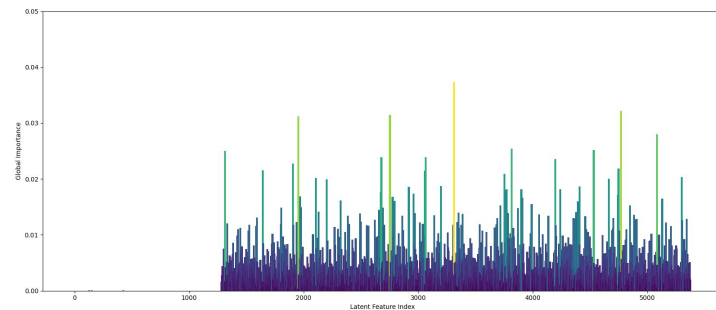
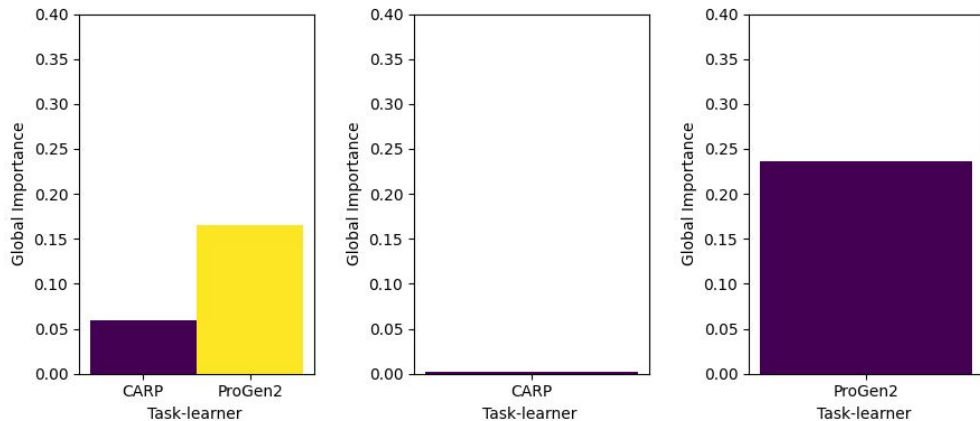
Results

Experiment 4 – Feature-Level pLM Combination Analysis

CARP & ProGen2

Exclusive reliance on ProGen2

⇒ Overshadowing latent spaces



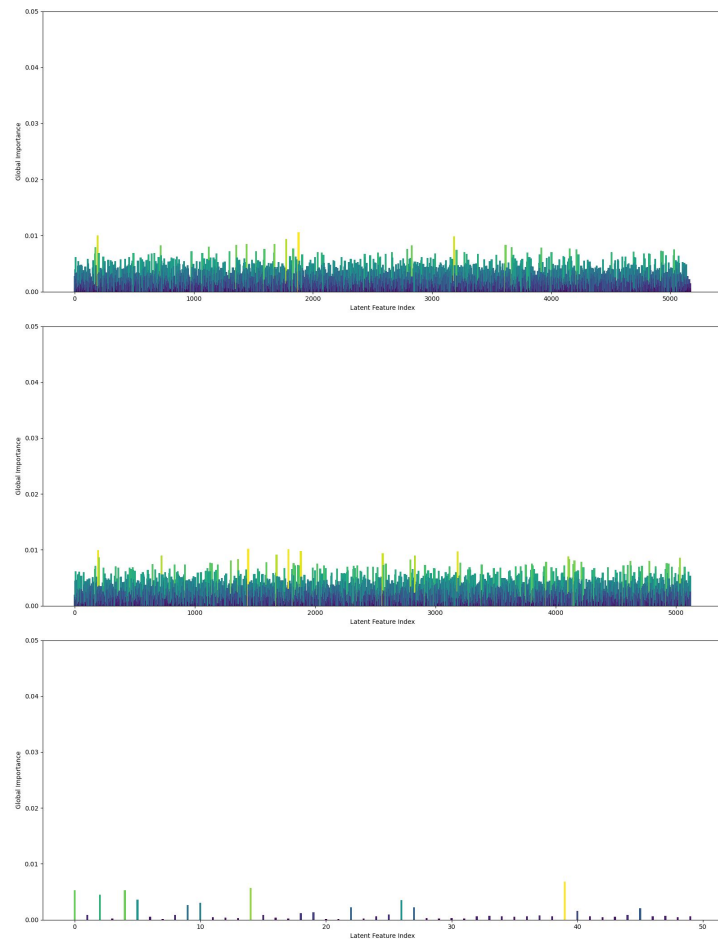
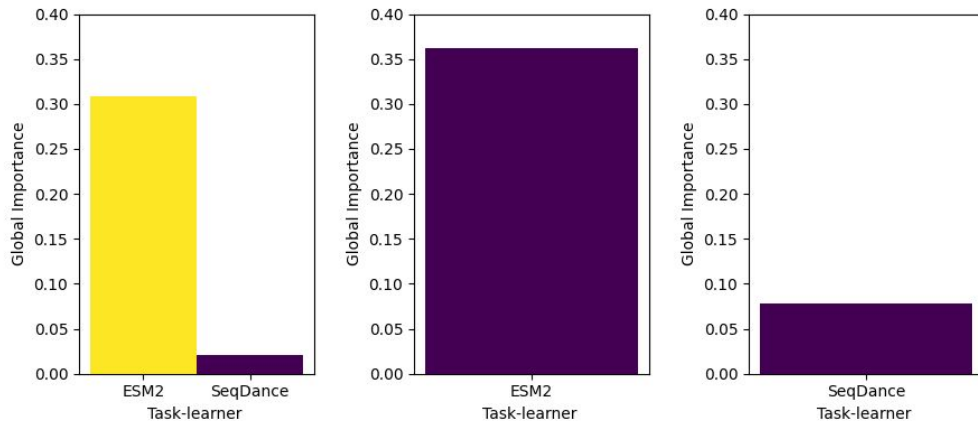
Results

Experiment 4 – Feature-Level pLM Combination Analysis

ESM-2 & SeqDance

Change in #important features

⇒ Complementing latent spaces



Conclusion

1. **Difference** wild-type & variant combination strategy is better because:
 - ⇒ It **does not double** the input space **dimensionality**
 - ⇒ Lets the MLPs **focus exclusively on the mutation**, not the context
2. **Greater #parameters & embedding dimensionality $\neq$ better pLM**
 - ⇒ **Architecture and training objective** (, fine-tuning, ...) matter more
3. **RF** showed **better class separation**, while **ML** had the **best accuracies** and F1-scores
 - ⇒ All tested architectures showed **advantages and disadvantages**
4. **pLM latent spaces can overshadow, complement and overlap**

Future Work

- Per-residue Embeddings
- Filtering out unimportant latent features
- pLM benchmarking frameworks/datasets (e.g. FLIP, TAPE)
- Comparison with other pLM ensembles (e.g. VarDCL)

Summary

Research Training

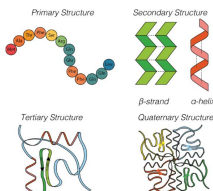
Background – Biology

Protein **function** defined by **conformation**

⇒ Protein dynamics = constant microscopic motion

Many types of mutations

⇒ Mutations may or may not affect function



(Morris R et al., 2022)

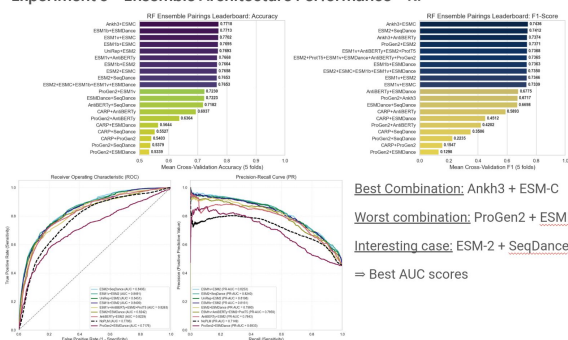
	No mutation	Silent mutation	Missense mutation	Nonsense mutation	Frameshift mutation
DNA	GAA TGC CCG	GAA TTT CCG	GAA TGC CCG	GAA ATC CCG	GAA ATG CCG
mRNA	CUU AAC GCA	CUU AAA GCA	CUU AAG GCA	CUU UAG GCA	CUU UAA GAG
protein	Glu Lys Gly	Glu Lys Gly	Glu Arg Gly	Glu STOP	Glu Asn Glu

(Morris R et al., 2022)

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Results

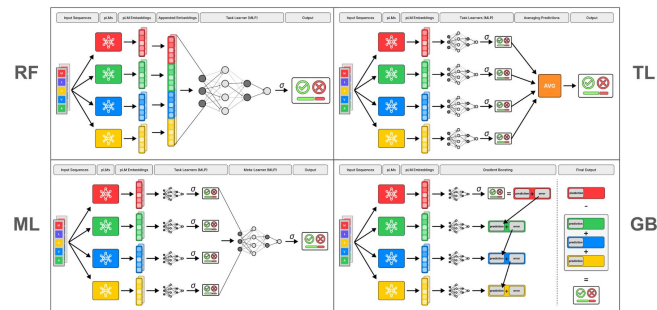
Experiment 3 – Ensemble Architecture Performance – RF



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Methods

Ensemble Architectures



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Conclusion

1. **Difference** wild-type & variant combination strategy is better because:
 - ⇒ It **does not double** the input space **dimensionality**
 - ⇒ Lets the MLPs **focus exclusively on the mutation**, not the context
2. Greater #parameters & embedding dimensionality ≠ better pLM
 - ⇒ **Architecture and training objective** (, fine-tuning, ...) matter more
3. **RF** showed **better class separation**, while **ML** had the **best accuracies** and F1-scores
 - ⇒ All tested architectures showed **advantages and disadvantages**
4. pLM latent spaces can **overshadow, complement and overlap**

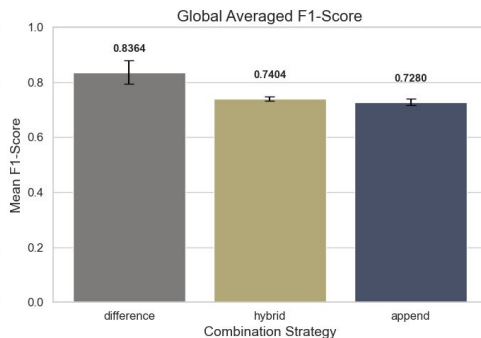
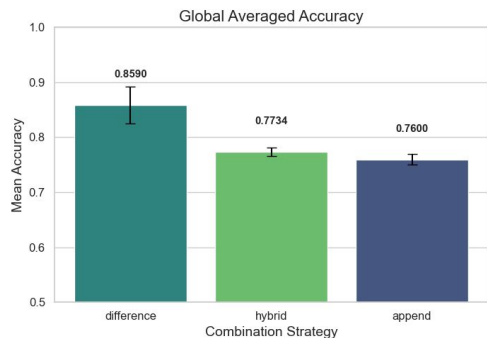
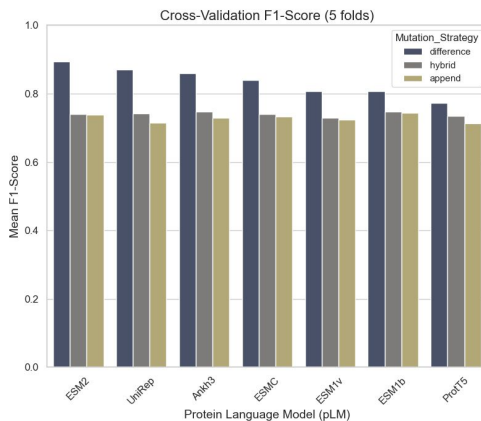
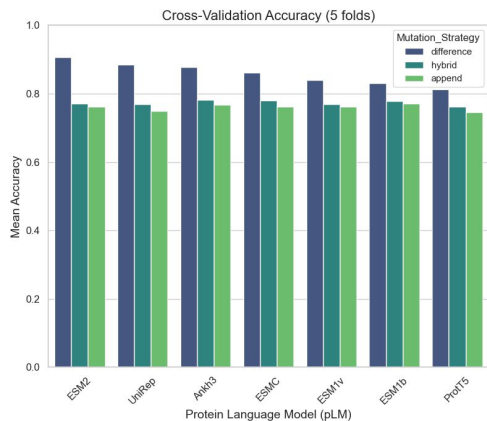
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References

- Morris R, Black KA, and Stollar EJ. Uncovering protein function: from classification to complexes. *Essays in Biochemistry* 2022 Aug; 66:255–85. doi: 10.1042/EBC20200108
- Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, Kaiser L, and Polosukhin I. Attention is all you need. *Advances in neural information processing systems* 2017; 30
- Chen S and Guo W. Auto-Encoders in Deep Learning—A Review with New Perspectives. *Mathematics* 2023; 11. doi: 10.3390/math11081777.
- Krizhevsky A, Sutskever I, and Hinton GE. ImageNet classification with deep convolutional neural networks. *Communications of the ACM* 2017; 60:84–90
- Schmidt RM. Recurrent neural networks (rnns): A gentle introduction and overview. *arXiv preprint arXiv:1912.05911*. 2019 Nov 23.
- Zhang H, Zheng G, Xu Z, Zhao H, Cai S, Huang Y, Zhou Z, Wei Y. VarDCL: A Multimodal PLM-Enhanced Framework for Missense Variant Effect Prediction via Self-distilled Contrastive Learning. *bioRxiv*. 2026 Mar 17:2026-03.
- New York Times, 2009. <https://archive.nytimes.com/bits.blogs.nytimes.com/2009/09/21/netflix-awards-1-million-prize-and-starts-a-new-contest/>
- Mienye ID, Sun Y. A survey of ensemble learning: Concepts, algorithms, applications, and prospects. *Ieee Access*. 2022 Sep 16;10:99129-49.

Results

Experiment 1 – Wild-Type & Variant Combination Strategies



Difference outperforms both

⇒ contains **explicit mutation information**

⇒ model **focuses on mutation**, not context

Append is the worst strategy

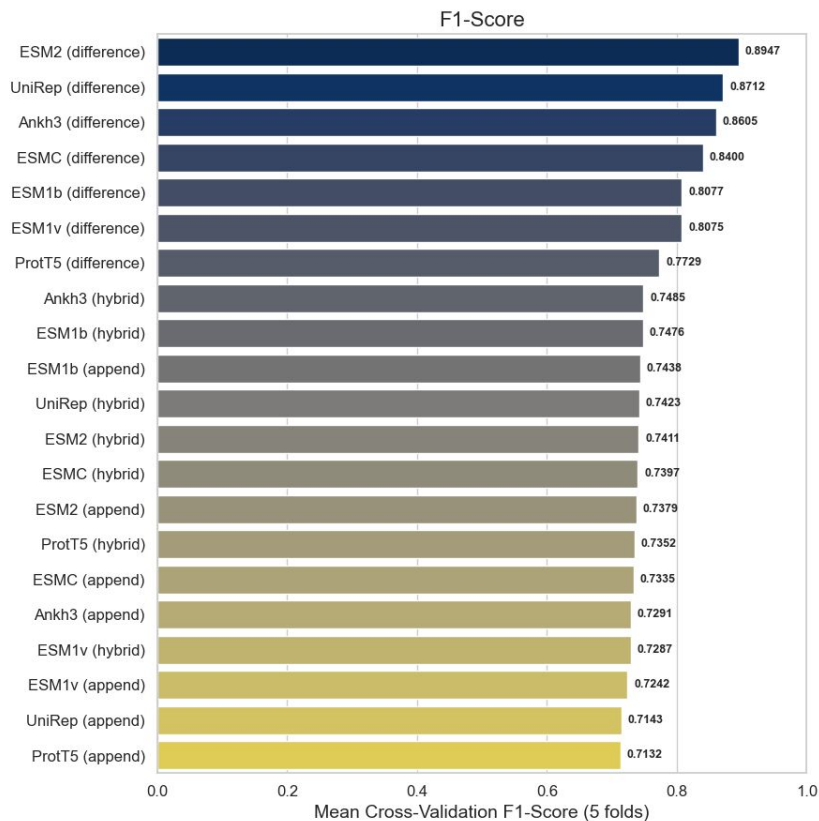
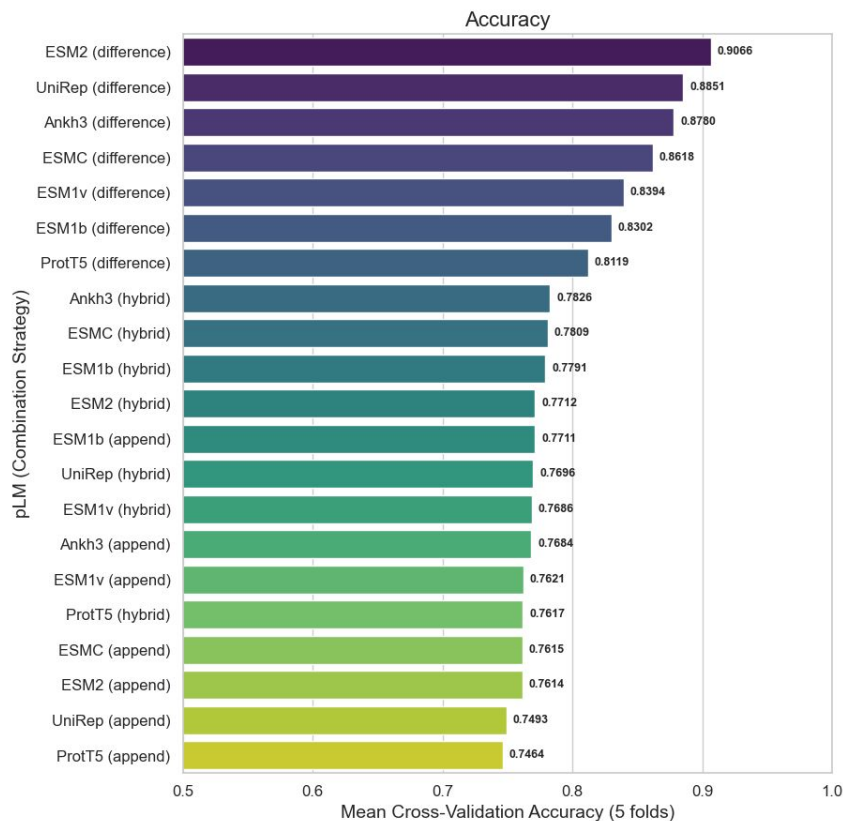
⇒ doubles input dimensionality

⇒ no explicit mutation information

Hybrid is slightly better than append

Results

Experiment 1 – Wild-Type & Variant Combination Strategies



Results

Experiment 2 – Individual pLM Performance

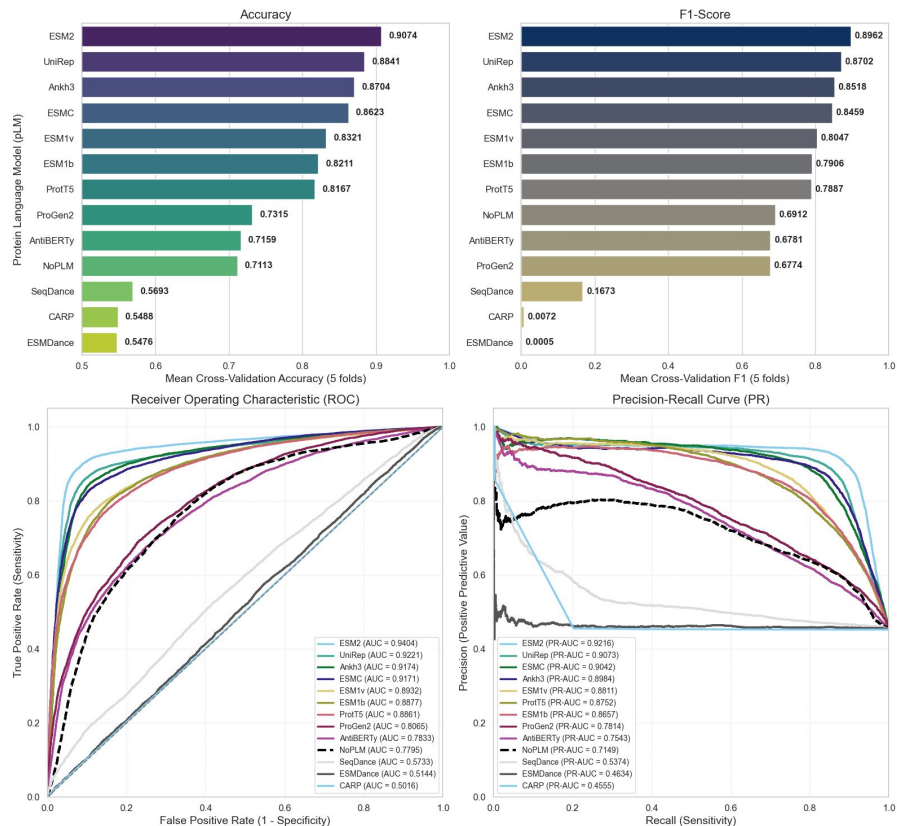
Larger models > smaller models

⇒ Except:

- ProGen2 (autoregressive)
- CARP (CNN)

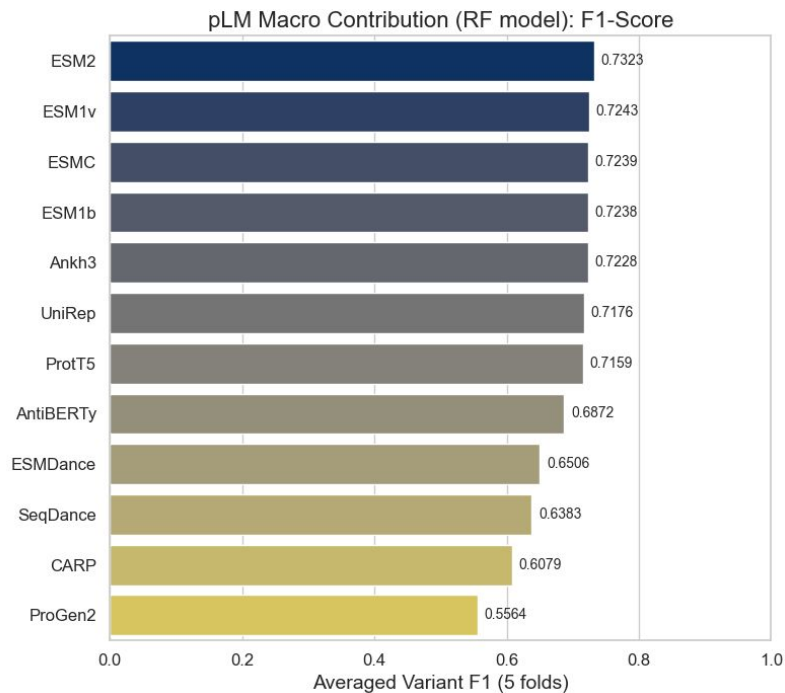
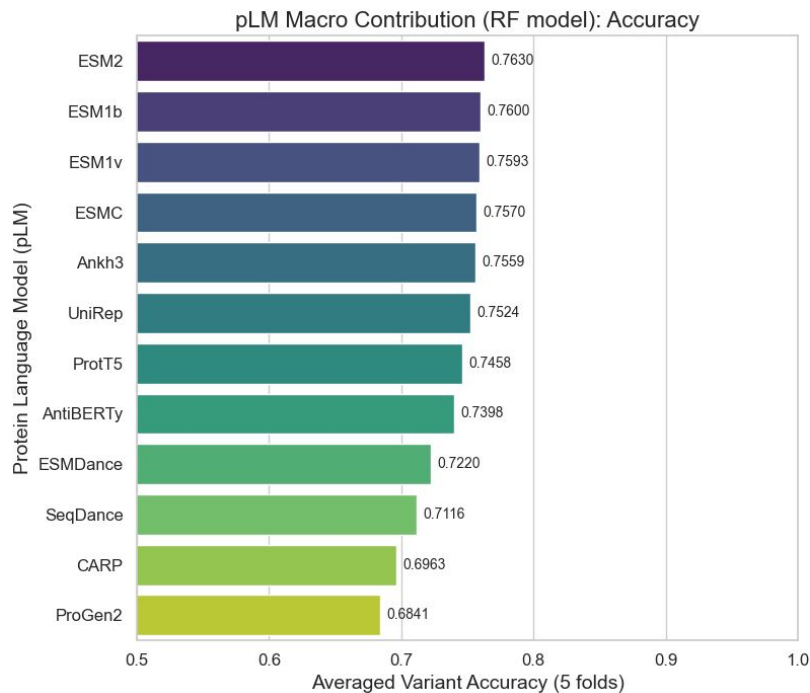
SeqDance, CARP, ESMDance < NoPLM

⇒ ESM-2 outperforms all other pLMs



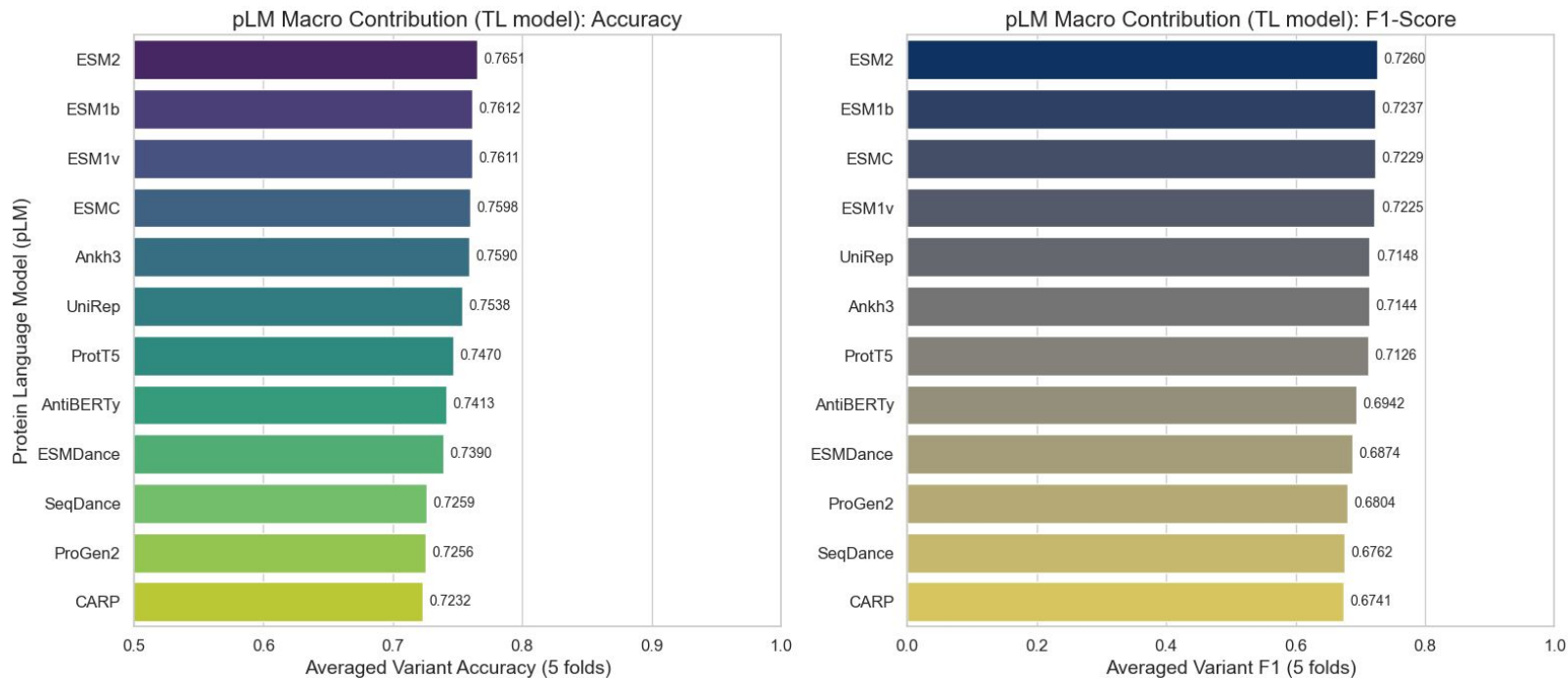
Results

Experiment 2 – Individual pLM Performance – Macro Contribution (RF)



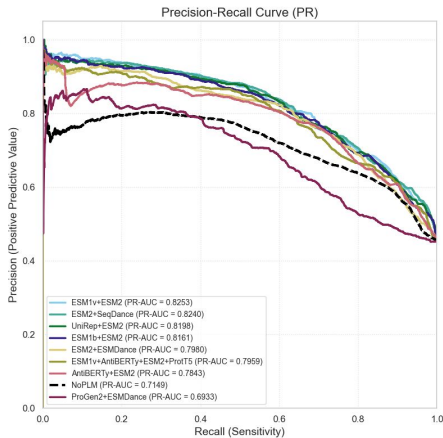
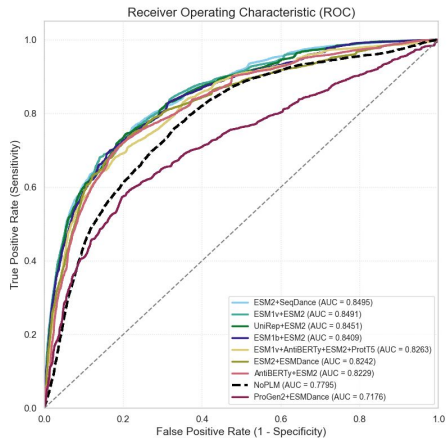
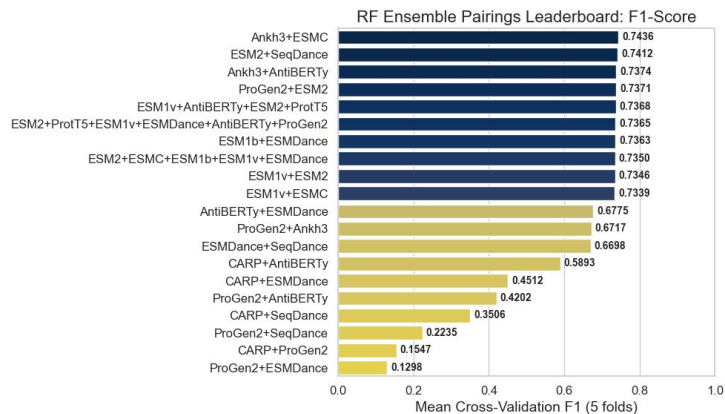
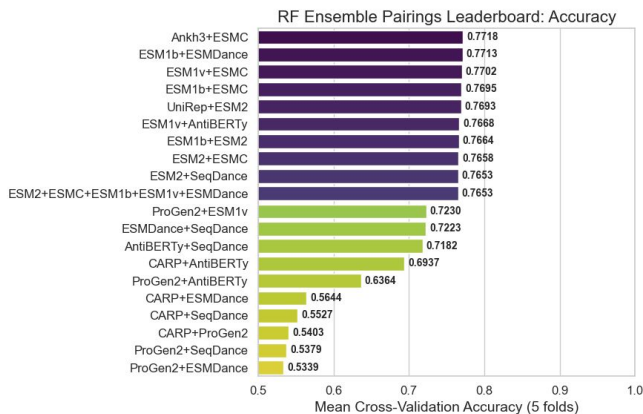
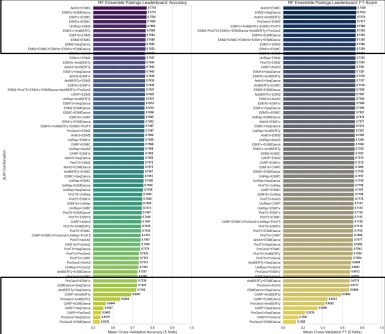
Results

Experiment 2 – Individual pLM Performance – Macro Contribution (TL)



Results

Experiment 3 – Ensemble Architecture Performance – RF



Best Combination: Ankh3 + ESM-C

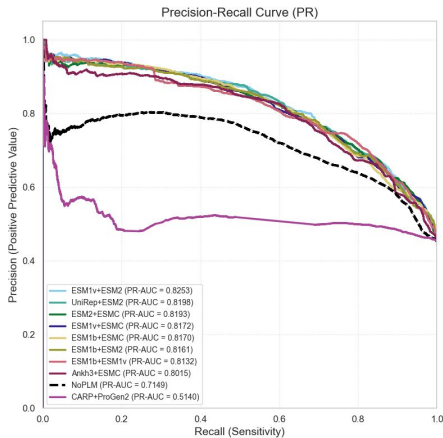
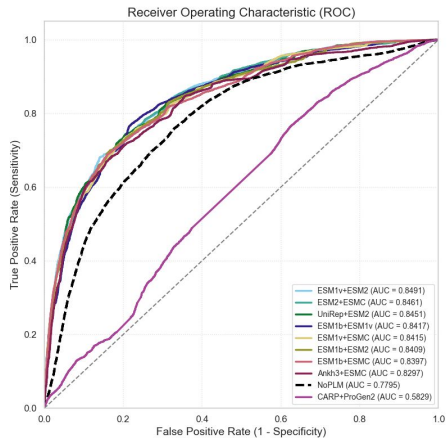
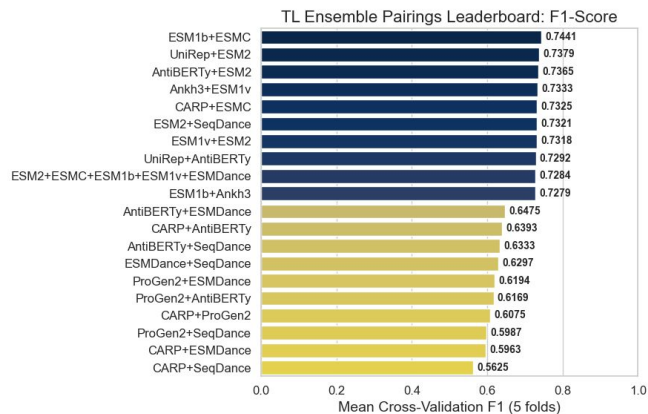
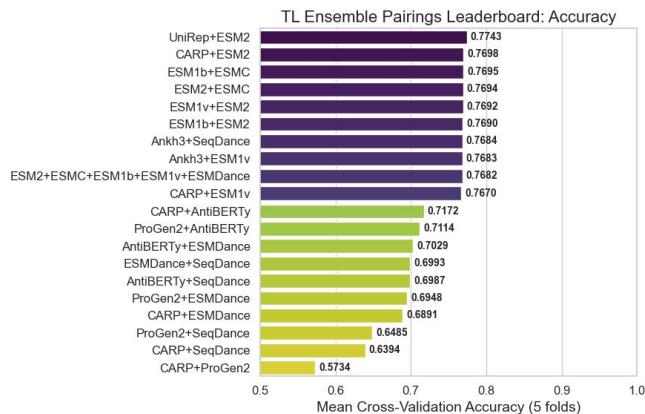
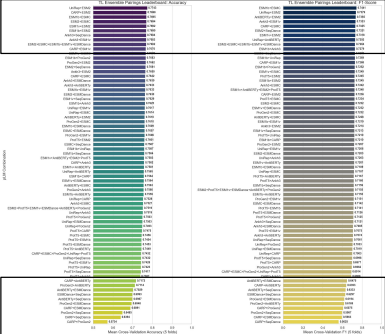
Worst combination: ProGen2 + ESMDance

Interesting case: ESM-2 + SeqDance

⇒ Best AUC scores

Results

Experiment 3 – Ensemble Architecture Performance – TL



Best Combination: UniRep + ESM-2

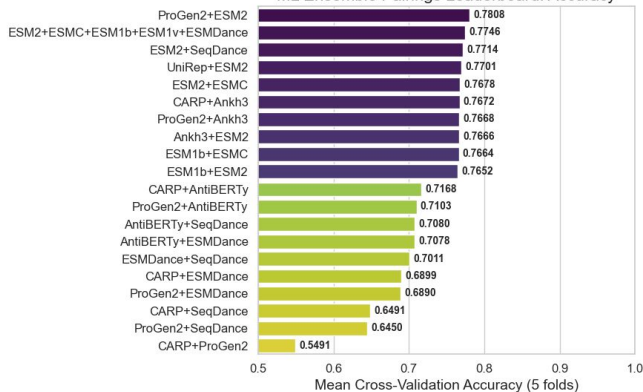
Worst combination: CARP + ProGen2

⇒ Accuracy/F1-score Lower bound is higher compared to RF

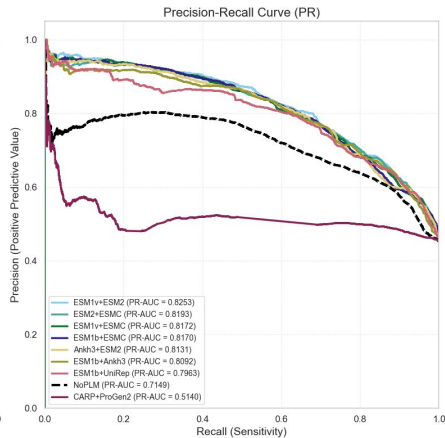
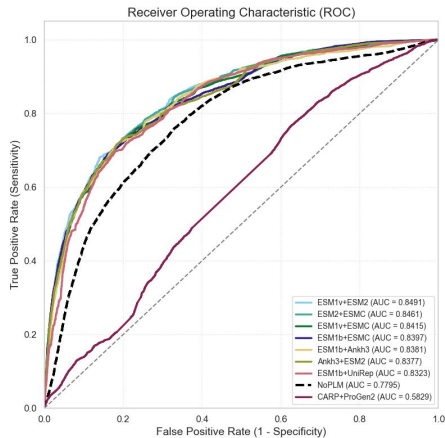
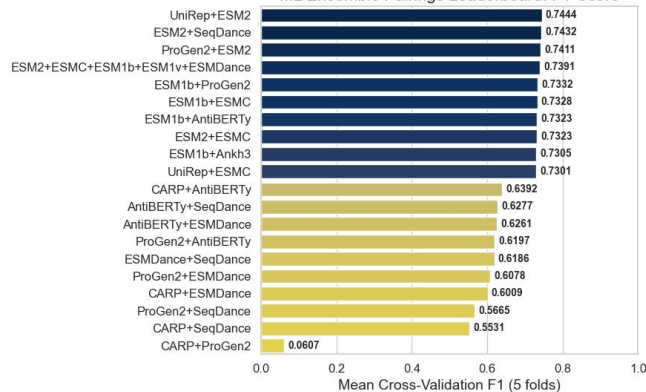
Results

Experiment 3 – Ensemble Architecture Performance – ML

ML Ensemble Pairings Leaderboard: Accuracy



ML Ensemble Pairings Leaderboard: F1-Score

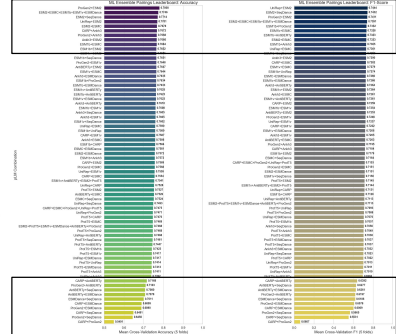


Best Combination: UniRep + ESM-2

Worst combination: CARP + ProGen2

⇒ Best accuracy/F1-score

(more expressive version of TL)

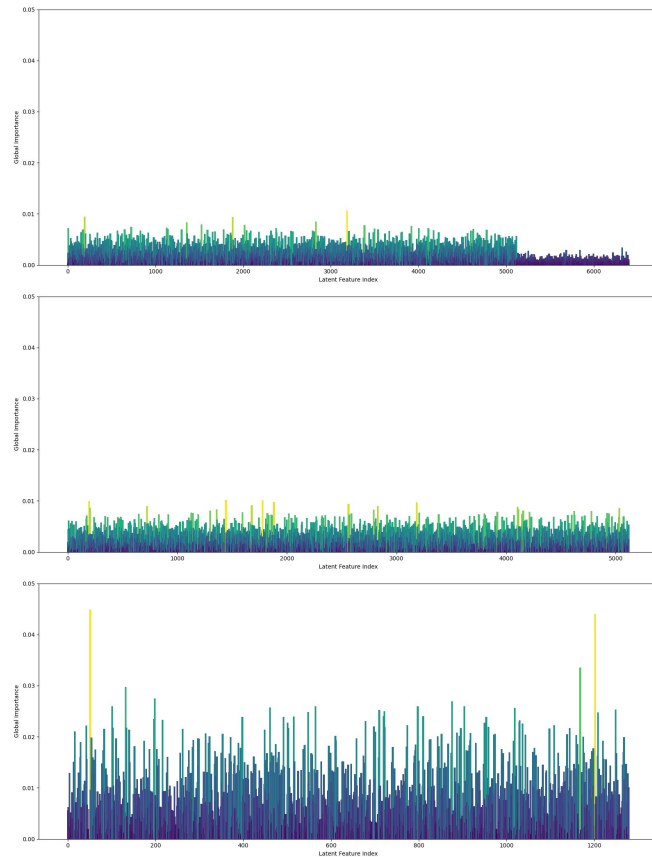
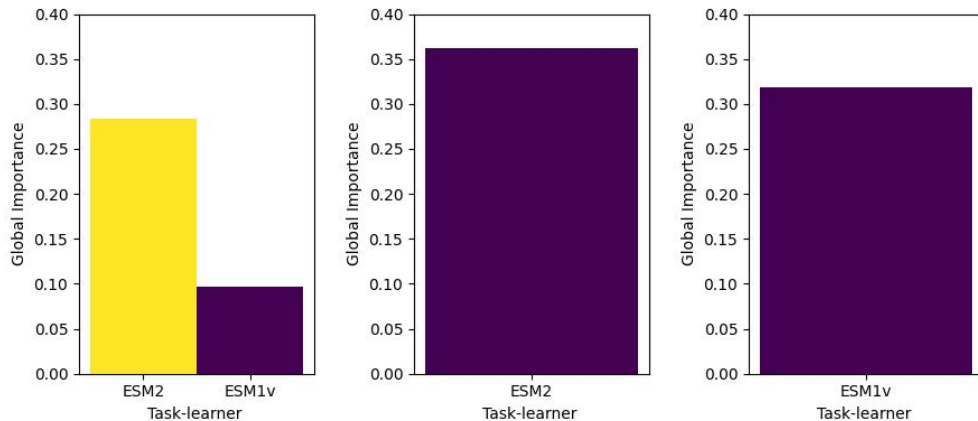


Results

Experiment 4 – Feature-Level pLM Combination Analysis

ESM-2 & ESM-1v

⇒ Overlapping latent spaces

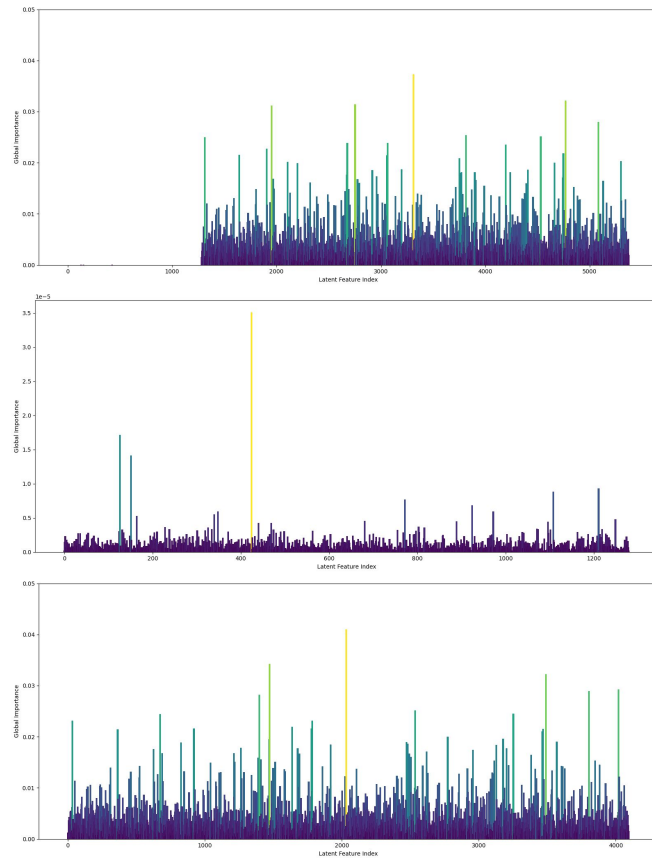
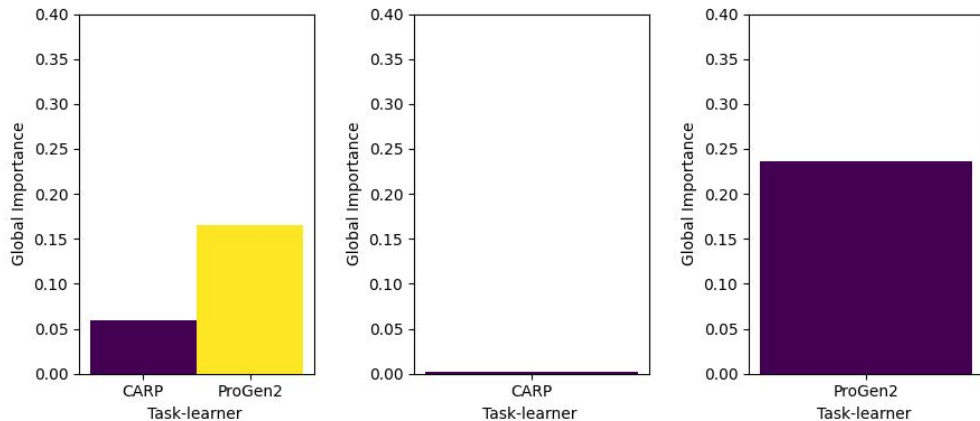


Results

Experiment 4 – Feature-Level pLM Combination Analysis

CARP & ProGen2

⇒ Exclusive reliance on ProGen2

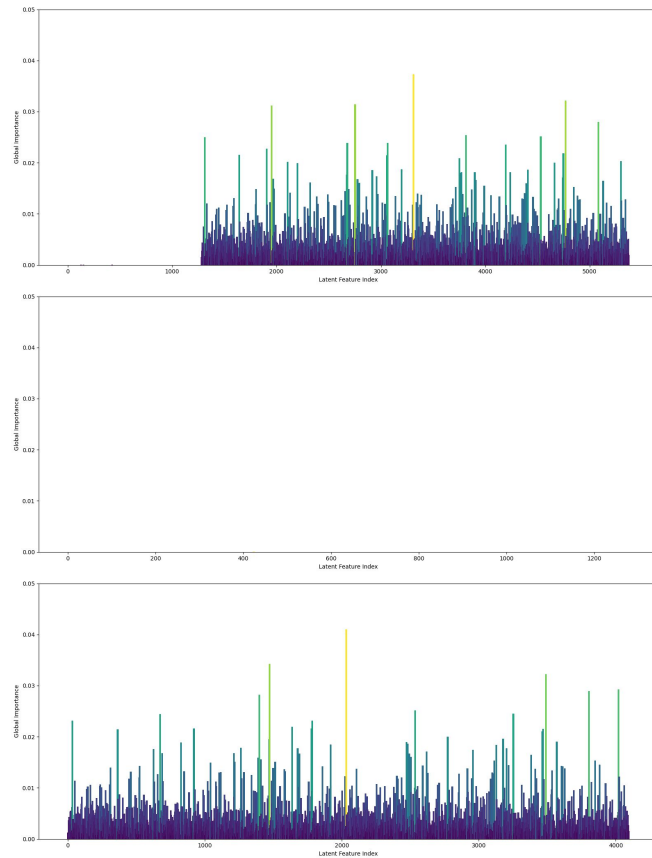
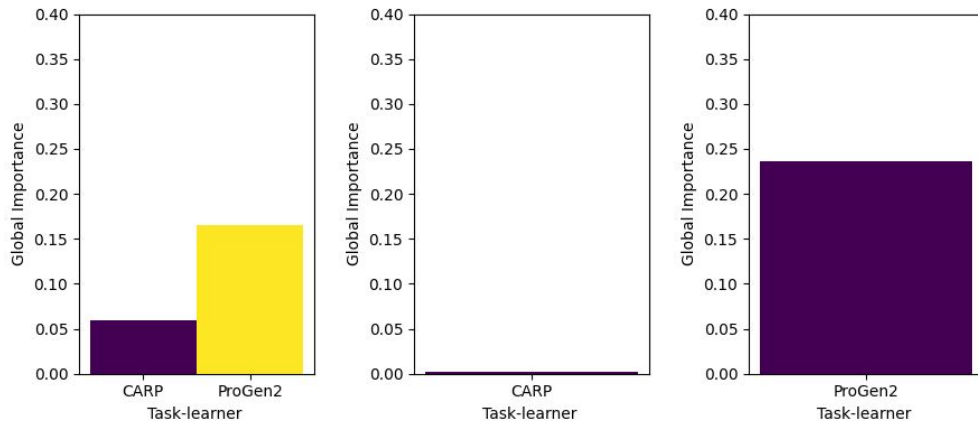


Results

Experiment 4 – Feature-Level pLM Combination Analysis

CARP & ProGen2

⇒ Exclusive reliance on ProGen2

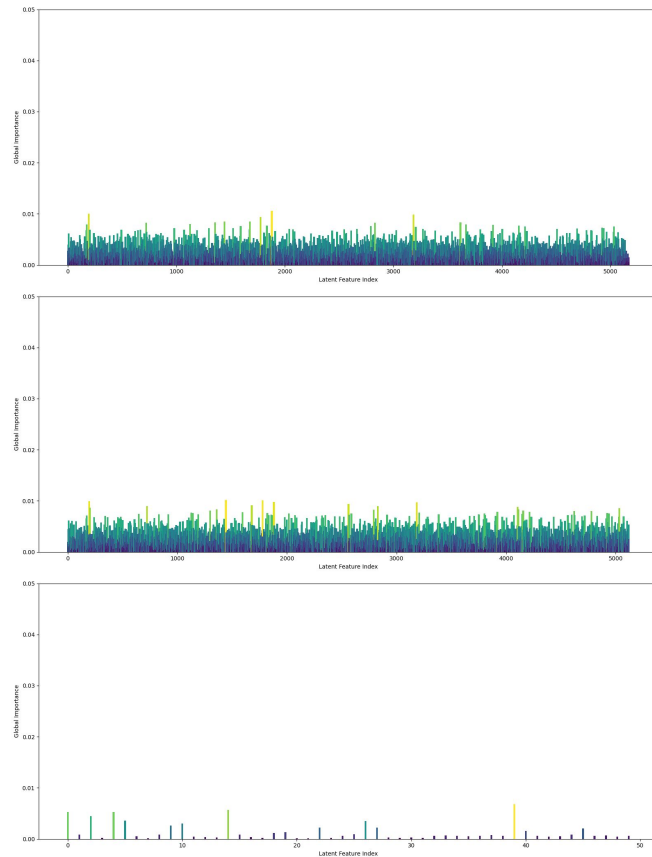
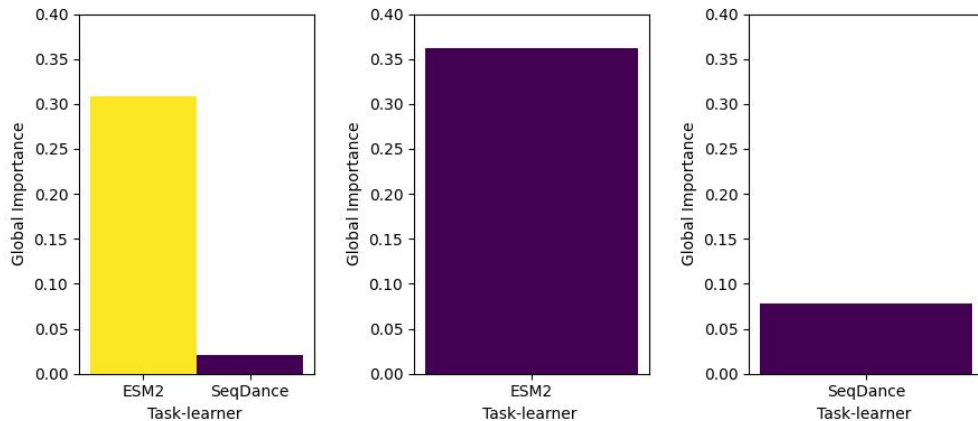


Results

Experiment 4 – Feature-Level pLM Combination Analysis

ESM-2 & SeqDance

⇒ Complementing latent spaces



Section

Subsection

Content